Slop Ninja

Public detectors, attested private transformation,
and paid inference on Bittensor

Slop Ninja Research

September 11, 2026
Draft 0.21

Abstract

Slop Ninja proposes a Bittensor subnet with two tasks: author and AI-origin detection (A), and author-directed text transformation (B). B jointly targets an authorized writer's style and evasion of Pangram and qualified subnet AI-origin detectors. Rewrites must retain the source's arguments, details and qualifications while removing unwanted formulaic writing. Every B benchmark measures both objectives; rewards require independent preservation, readability and requested-tone approval. Indistinguishability from the writer's own work remains a target for blinded evaluation.

A miners publish complete, immutable inference artifacts; B miners keep generation models private. B can run A locally for training and search, and validators independently reproduce A scores. Pangram initially anchors A's origin-detection quality target. An independently confirmed superior subnet detector replaces it; continuing comparisons let Pangram regain that role if it surpasses the incumbent. Detector results establish neither human authorship nor preserved meaning.

Miners qualify for the interfaces they commit to serve. A requires qualified public artifacts and fulfillment of assigned obligations as training requesters. B requires qualified attested generation and bounded rewrite service for authorized A training requests. Miners must meet availability requirements across all their committed roles. Emissions reward benchmark performance; miners separately sell asynchronous inference at market prices denominated in the subnet's alpha token. B transformation is the primary paid product, with optional paid A hosting.

For paid B jobs, customers encrypt inputs to a verified trusted execution environment (TEE), a hardware-protected runtime. Customers verify fresh evidence covering the execution code, CPU, GPU and transfer path before releasing input keys. Model owners independently verify the instance before releasing model keys; they receive no customer input keys and cannot authorize their release. Customer text is not automatically disclosed to validators, routing services or other miners. Customers may separately disclose evidence to a chosen validator. Every credited B candidate and paid B job needs an execution receipt: a record signed by the protected runtime that binds generated text to the registered private model version and serving profile.

On-chain contracts coordinate jobs and payments. Paid B settlement requires the execution receipt, complete publication of the customer-encrypted result, and a delivery certificate: validators' signed agreement that the evidence satisfies delivery rules. Customer acknowledgment is unnecessary. Delivery certificates and final benchmark judgments require strictly more than two-thirds of frozen eligible validator weight, assuming less than one-third is dishonest and every signer inspects the evidence. Delivery certification grants no decryption access and establishes no editorial quality.

This draft specifies the proposed protocol and the limits of its prototype. Detector accuracy, reliable cleanup in the requested voice, attested CPU/GPU inference and live reward settlement remain to be independently demonstrated.

1 Purpose and scope

Slop Ninja's transformation mechanism jointly targets author-style matching and evasion of Pangram and qualified subnet AI-origin detectors. Both are primary objectives of B. The aim is to remove unwanted generic and mechanically patterned writing while preserving the source's full argumentation, details and qualifications. The result should read naturally in a particular authorized writer's voice, with their characteristic word choices, grammatical constructions and rhetorical habits and the tonal intent of the requested style. Here, *slop* means unwanted filler, vague or formulaic phrasing, needless repetition and other patterns that weaken the intended writing. Intentional informality, unusual constructions and recognizable personal habits belong to the target profile when appropriate to the brief.

The product thesis is that readers object to slop because it combines weak writing with a recognizable generic model voice. Removing those defects should produce good writing that sounds like its author. Successful cleanup therefore means the result has ceased to be slop under this definition. Reader judgments of the writing and author fit must test that thesis alongside the required detector benchmark.

The requested writer's profile supplies a direction for lexical, grammatical and rhetorical changes. Adversarial detector feedback adds pressure to remove patterns recognized as model-generated. B uses both signals to guide the same rewrite. Their practical reinforcement is a testable hypothesis; author fit and detector evasion each retain their own measurements.

The desired endpoint is a revision that readers familiar with the writer cannot reliably distinguish from that writer's own comparable work, while the required substance remains intact and the unwanted patterns have been removed. This is an empirical target. Evaluate it through blinded comparisons in the same register, independent tests on held-out author samples, and full preservation review. Neither a favorable detector score nor proximity to a learned profile alone can establish that result. Writers may also request greater accessibility or rhetorical force; those changes must respect the source and brief.

Illustrative revision. The following brief and passages are synthetic. The brief summarizes a fictional target writer's profile; actual benchmark tasks require independent authorized reference writing.

Brief: Write a team update in the project lead's voice: short sentences, contractions and a calm tone. State uncertainty directly and retain the reason for the decision, every quantity and the unresolved question.

Source: "Following a review of the pilot, the team has made the decision to extend testing by two weeks. Three of the twelve participants reported missing notifications. This extension may help us establish whether the problem also affects new accounts; we do not yet know."

Revision: "We've reviewed the pilot and decided to extend testing by two weeks. Three of twelve participants reported missing notifications. We don't yet know whether new accounts are affected. The extra testing may help us find out."

The revision removes the long decision phrase and follows the requested voice while retaining the reports, quantities and uncertainty. Replacing "may help" with "will establish" would fail preservation. This example illustrates the editorial judgment; it has no measured author-fit or detector result and establishes no benchmark pass.

The model-publication split is intentional. A should make strong author/origin detection available for anyone to run and use in training, including external detector providers. Matching Pangram-level origin detection is an evaluation target, not an achieved result. Pangram initially sets A's origin-detection quality target. An eligible subnet detector takes over after confirmed superiority, and Pangram becomes the target again if it later outperforms the incumbent. Continue comparing both against independent production records; author identification retains its own known-author labels. Public A models give B immediate local feedback while B miners retain their transformation models. A submissions must permit free artifact access, local inference and reuse for training private B models. Hosting still costs compute and may be sold separately. The subnet aims to make detector access a shared resource while miners compete to provide useful private transformation services. Both B emission eligibility and paid customer service require qualified attested execution. Customer jobs use attested private inference under Section 6.3. Customers verify the execution policy and instance before disclosing text, while B miners retain the model they sell access to. Hardware and audited software become explicit trust dependencies; attestation does not certify editorial quality.

The initial subnet evaluates two tasks separately:

A. Identify authors and documented human/model production histories.
B. Clean up unwanted writing patterns and match an authorized writer's profile while evading Pangram and qualified subnet AI-origin detectors, preserving meaning, detail and target tonal intent.

The detector component of B's strict success target requires the preservation and quality checks, a score strictly below 10% on Pangram's reported AI-generated **plus AI-assisted** fraction, a majority pass against a common panel of three qualified subnet origin detectors, and a pass against the strongest qualified prior-round detector at its calibrated threshold. The panel's lead detector is selected by held-out origin Brier performance; A's quality anchor is selected by sensitivity at fixed human false-positive operating points and may be a different detector. The panel contains its lead and two hash-selected qualified detectors, frozen for all B candidates in a matched batch (Section 8.2.3). Reaching zero is a desirable measured outcome, not a promise of this design. Model-origin text retains its recorded origin after a successful rewrite. Detector evasion does not change its production history.

The adversarial loop connects these two tasks: detection miners learn to recognize model-generated revisions, while transformation miners try to defeat qualified detectors from completed rounds. The benchmark fixes the opponent-assignment and scoring rules and retains source provenance and each candidate's production record as the models improve.

The initial scope is English prose, document-level author/origin predictions, and bounded asynchronous editing jobs. Span-level origin labels, multilingual rewards, and arbitrary-length manuscripts need their own validated datasets. There is no requirement for token streaming or interactive latency. Writer-directed cleanup belongs to B; there is no separate writing-improvement mechanism at launch. Audience and editorial controls remain part of B's briefs and preservation review.

2 Related subnets, services, and research

This representative survey covers direct task competitors and relevant evaluation and inference infrastructure. It uses project-maintained documentation and research publications checked on September 9–11, 2026. Subnet numbers below follow the projects' own documentation; this is not an independent audit of live chain state. Published service claims are distinguished from results measured by this project.

2.1 Bittensor precedents

It's AI and BitMind's GAS supply the closest task and incentive precedents. It's AI documents human/AI text classification on SN32, with miners serving detection requests. Its FAQ says the team does not know which models its leading miners use. This provides a precedent for evaluating detection endpoints with undisclosed implementations. GAS documents an explicit competition between detection models and generators that try to fool them [1–3].

Project	Documented approach	Relevance to Slop Ninja
It's AI (SN32)	Human/AI text detection, including sentence-level probabilities, through miner endpoints [1,2]	Direct comparison for origin detection; compare calibration and generalization on the same independently labeled text
BitMind GAS (SN34)	Image/video/audio detector submissions and image/video generator endpoints; generation rewards include fooling detectors [3,4]	Direct precedent for adversarial incentives; text revision adds source-preservation and requested-voice constraints
Macrocosmos Finetuning (SN37)	Miners publish competition models on public Hugging Face repositories for validator evaluation [5]	Precedent for rewarding model improvements through submitted weights; Slop Ninja pairs public A artifacts with attested private B serving
Chutes (SN64)	Serverless model inference and metered service, with documented confidential-serving options [6,7]	Paid inference and private serving infrastructure; writing quality requires separate task evaluation
Targon (SN4)	Managed compute and inference; confidential VMs with attestation-bound key release for proprietary workloads [8,9]	Hardware-based endpoint protection and private-model execution are established design topics

These projects establish that detection competition, adversarial generation, language-model training rewards, and paid private inference already have Bittensor precedents. Their documented mechanisms evaluate different properties. GAS requires submitted detector weights; Slop Ninja also requires public A artifacts, paired with private B generation and independent verification of B's self-scores.

2.2 Confidential inference services

Privatemode documents client-side verification of measured workload policies before key exchange, combined CPU/GPU attestation, and explicit prompt-cache isolation [10,11]. Chutes documents confidential serving and request/response encryption, and Targon describes hardware attestation and key release. Slop Ninja proposes to adopt that hardware-based direction for all paid B jobs, with client verification of the complete application and private-model binding before key release. Provider support alone does not establish that our runner or customer protocol is secure. Protected execution also does not establish that a revision preserved an argument. The proposed association between private model versions, benchmark results and paid service requires the instance-specific key-release policy in Section 6.3; we do not assume that a provider's existing key-distribution policy already enforces it.

2.3 Existing writing services

Pangram and GPTZero already offer human/model-origin classification, including mixed or assisted categories. Pangram is the required external opponent here; GPTZero is another useful comparison on rights-cleared benchmark material. Their output categories and scores must be mapped explicitly to the evaluation's documented production labels. Agreement among detectors does not establish an author's identity [12,13].

Grammarly documents personal voice profiles and a feature for rewriting in the user's voice. Wordtune offers sentence and paragraph rewriting, formality changes, shortening, and expansion. QuillBot documents multiple paraphrasing modes, including custom and Humanizer modes, and warns that its Creative mode can alter meaning. These are direct comparators for author-directed revision and its editorial controls; personalization and editorial controls are existing product capabilities [14–16].

Undetectable AI explicitly markets humanization modes intended to bypass detectors. That documents an existing commercial objective; this survey does not independently establish its bypass or preservation performance [17]. Comparing such services requires the same sources, briefs, retained outputs, and blinded preservation review used for subnet miners. Advertised detector scores cannot substitute for that comparison. Hosted service comparisons use authorized benchmark text and do not create an exception to confidential customer delivery.

2.4 Adoption and demand

Model providers and publishing platforms are bringing provenance checks into ordinary writing workflows. Anthropic announced text watermarking on August 14, 2026. Its coverage documentation says Claude models launched on or after August 2 receive marking at launch, while support for older models remains in rollout. Supported outputs are marked worldwide across Claude products, the API and partner platforms [18,19]. Google announced SynthID text watermarking in the Gemini app and web experience in May 2024 [20]. Substack now offers Pangram-powered reader scans for posts and notes published on or after July 21, 2026, with author controls to disable scanning on individual items [21].

These deployments support a market thesis for both subnet tasks. Publishers and readers may want affordable, independently evaluated detection as provenance checks become more common. Writers using model assistance may want stronger control over their own voice and a way to remove unwanted generic phrasing without losing substance. Public A models support accessible detection and local evaluation. Private B transformation is the primary paid product, with attested inference for every paid B job and optional A hosting as a secondary service. We expect broader provenance checking to increase demand for these capabilities. The size of that increase, willingness to pay and repeat use remain to be measured through customer pilots.

Anthropic distinguishes keyed watermark detection from classifiers such as Pangram that infer origin from linguistic patterns. Its watermark detector is in private preview for eligible organizations [18]. Passing Pangram and public A models therefore does not establish removal of a Claude watermark. Watermark detectability, inferred origin, prose quality and author fit require separate measurements. The proposed B product remains cleanup in the requested writer's voice; any future watermark benchmark would need access to the relevant detector and its own evaluation protocol.

2.5 Research foundations and proposed contribution

Gero et al. model style through controls over function-word frequencies and syntactic constructions while retaining content words. This is a direct antecedent for combining lexical and grammatical features. StyleRemix uses interpretable style axes and LoRA modules for authorship obfuscation, providing a relevant comparison for controllable rewriting [22,23]. Authorship obfuscation changes evidence available to an author classifier; matching a requested writer and preserving every required claim remain separate evaluation questions.

The PAN/ELOQUENT 2024 task paired human/AI detection with generation and obfuscation intended to avoid it, offering a prior benchmark for an adversarial loop. DIPPER demonstrated paraphrase attacks on several then-current detectors and studied a retrieval-based defense. Those results motivate adaptive opponents and held-out evaluation; they do not establish performance against Pangram 4 or Slop Ninja's future incumbents [24,25].

The proposed contribution is a text-focused protocol connecting origin-detection competition to cleanup in a particular writer's voice, with independent preservation, readability, and requested-voice review. The transformation target combines removal of unwanted writing patterns with retention of the writer's individual habits and required evasion of Pangram and qualified subnet detectors; retired, authorized rewrites supply later detector-training examples, with source families kept separate from private evaluation. A artifacts are public; B miners retain their private transformation models. Both can sell hosted inference through an alpha-denominated market. Attested service binds customer key release to the accepted execution boundary. This combination requires validation against the precedents above. The survey establishes neither uniqueness across all subnets nor superior performance by an implementation that has not yet been built.

3 Participants and the two kinds of work

A miners publish immutable, complete detector artifacts. B miners may retain private transformation models and register signed rewrite endpoints, with qualified attested instances required for B emissions and paid work. Both register capabilities and authenticated protocol identities. A miners may separately offer paid detector hosting. Customers buy detection and transformation jobs through separately quoted offers. Validators execute the frozen A artifacts for authoritative benchmark scores, evaluate B outputs, and submit weights. For paid B, they verify TEE receipts and complete encrypted delivery without reading customer text. Benchmark curators maintain permitted source material, production records, and hidden evaluation labels. Customer-facing gateways may relay ciphertext, but receive no decryption key.

Benchmark operator and coordinator denote administrative duties assigned to named curators and validators before each round: funding and managing the round, recording protocol-derived assignments, and obtaining baseline evidence. A pilot may combine these duties in one service. The task owner is the customer for private inference and the curator for a benchmark. Authorized evaluators and auditors are assigned validators; judges, readers, and authorized reviewers perform independent review for those validators. Automated judges require the qualification described in Section 8.3. The settlement adapter is the proposed contract software for customer escrow and alpha transfers, described in Section 7.

An *evidence preparer* is a hash-assigned validator who assembles the review dossier. Each *certifying validator* belongs to the round's frozen eligible set and independently checks the evidence. In certificate rules, a *signer* is such a validator signing a particular final verdict. These duties may overlap; preparation alone grants no approval authority. Section 10 fixes the weighted quorum.

An A *model-credit entry* records performance credit for one detector's inference content after exact duplicates are coalesced under Section 4.1. It uses the canonical submitting UID and existing A payout policy; it creates no additional UID, token or payout pool.

Property	Hosted customer inference	Subnet benchmark
Purpose	Serve a customer's request	Measure miners for emissions
Launch services	A: detection; B: transformation	A: detection; B: transformation
Source of text	Customer	Authorized benchmark collection
Default input recipients	B: customer and attested instance. A: operator too if ordinary hosting is selected	Curator, assigned service miners, validators, commissioned reviewers
Evaluation	Customer judges quality; validators certify paid B execution and encrypted delivery	Validator execution of frozen A artifacts; independent B output review and required attested receipts
Pangram submission	Excluded from the confidential job protocol	Required for every B benchmark
Payment	Customer's quoted fee	Performance-dependent emissions
Training reuse	No automatic reuse	Only under the collection's release terms

Publishing an A artifact does not grant its publisher access to private benchmark text. Validators execute it on authorized evaluation inputs; miners receive only the inputs required for their assigned tasks.

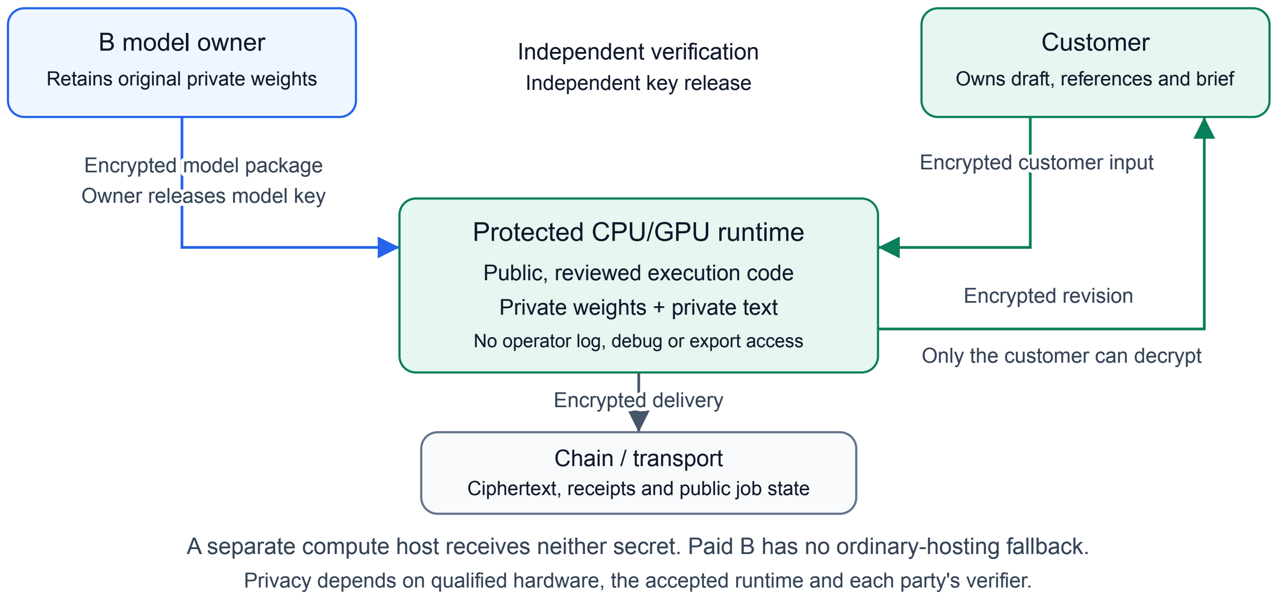

Figure 1: Private B inference. The model owner and customer independently verify the runtime before provisioning their secrets. Only that runtime processes both in plaintext; the customer receives the encrypted result. Validators check the receipt and complete ciphertext without keys. A separate customer disclosure can authorize editorial inspection.

Customer tasks do not become benchmark examples automatically. Validators must not obtain customer inputs through an audit exception, a shared support key, or an arbitration process. Any customer may initiate a separate, explicitly addressed evidence disclosure as described in Section 6.6. This applies to both detection and transformation requests. An author's uploaded reference collection can be more identifying than the query itself and receives the same protection.

Miners may advertise specialties, prices, length limits, and delivery times. Mandatory training tickets assign an A requester to a B rewrite provider on authorized text. A contributes published artifacts; B runs them locally for training and submits its benchmark self-scores for independent validator replay. A operates no required scoring endpoint. The network evaluates observable service performance and published A computation. An attested B receipt additionally associates an output with its private model version under the accepted hardware and runtime policy. It cannot establish training effort, ownership of every component or independent ownership of UIDs. Ordinary private generation is research-only. Emissions create an incentive to improve training; they do not reimburse unverifiable training claims.

4 Model interfaces and starter architectures

The two launch interfaces, A for detection and B for transformation, define compatible jobs and outputs. Miners may change their internal architectures, grammatical representations, search methods, and training data without changing those interfaces. Sharing a foundation model between tasks is permitted. A detectors must be published as complete immutable artifacts; an update enters a later qualification round. B model weights remain private. Attested B serving uses an audited, measured runtime with private weights loaded as data, as specified below. A training recipes and data need not be public, subject to the publication rights required for the resulting artifact. Those terms must permit free access, local inference and reuse for training private B models; an artifact available only behind a paid inference API does not qualify. Optional A hosting may still charge for compute and delivery.

4.1 Author and origin detection

An author task supplies a query and reference works for a gallery of candidate authors. The response contains probabilities over the supplied author IDs. The starter compares full sparse word/grammar profiles; a learned text encoder with an author embedding head is the next architectural comparison. Reference prototypes support new customer authors without fitting a permanent classifier for each identity. Unknown-author rejection requires a separate calibrated output and withheld-author evaluation.

An origin task returns probabilities over documented production histories: human-only, model-only, and mixed human/model work. The production record includes revision order and assistance type. Rewriting a human-only source with a model produces mixed human/model work; a model-to-model rewrite remains model-only under this initial label scheme. Pangram supplies a comparison, not the training label. The proposed encoder uses a separate origin head; author similarity must not be substituted for provenance.

ModernBERT-base [26], a 149M encoder, is a concrete starter candidate. Selection must compare it with the existing metric on new authors, topics, and registers. The subnet does not require that checkpoint or assume it will win.

Published artifact and execution policy. Each A artifact binds weights, features, tokenizer, preprocessing and postprocessing, calibration, reference pooling, executable dependencies, runtime, numerical precision, hardware requirements and output rules. A miners pin prompts and random generators where used. During qualification, validators establish a reproducible reference execution, including fixed tolerances and a resolution rule for numerical boundary cases. They validate and cache the qualified panel, its artifact hashes and thresholds before the curator reveals the B task or permits candidate generation. Validators execute that panel themselves for authoritative benchmark probabilities; a miner's reported score is not the scoring oracle.

The panel uses distinct UIDs and distinct inference-content hashes, excluding owner and signature metadata from the content identity. Identical content receives one model-credit entry and one possible panel seat; the earliest finalized accepted commitment wins, with canonical UID order for ties. Small changes and equivalent implementations remain a qualification problem.

For registration, A miners submit a domain-separated hash commitment binding their registration identity, round, canonical content digest and a fresh random 256-bit nonce. The nonce and registration fields belong to the commitment, not the inference-content identity; the nonce remains secret until opening. For previously unreleased content, miners wait for commitment finality before disclosing the digest, manifest or retrievable artifact. The round fixes a publication deadline before qualification and panel selection. By that deadline, the miner must open the commitment and provide the complete artifact; validators check the opening, content digest and qualification. Only accepted entries retain their original commitment's finalization priority. Missing or invalid publication reserves no credit, panel seat or priority in later rounds. Already-public artifacts and foundation-model components remain admissible. This ordering allocates duplicate credit; it does not establish who authored or first trained a model.

The benchmark scheduler selects reserve artifacts before task issue. After issue, a runner outage permits execution of the same artifact on a reserved runner, not substitution of another detector or threshold. An unresolved reference execution withholds credit under the candidate-specific or shared-evidence closure rules; it is neither B nonresponse nor an automatic pass.

Validators qualify each artifact against independent hidden author and production labels. They include targeted-trigger controls: a published model can faithfully execute a rule designed to favor a colluding B. Publication verifies computation, not reliable detection or independent ownership. The isolated runner provides no network or undeclared state and excludes miner identities and assignment metadata from model inputs; text itself can still reveal task roles, so this does not prove blindness.

4.2 Author transformation and detector evasion

Inputs are the source, independent target-author samples, requested style changes, and a preservation brief. The brief identifies unwanted writing patterns to remove and intentional voice features to retain. Before generation, it fixes the target tonal intent using any requested style and supplied reference samples. Dramatic authorial-style shifts can change tone while preserving the source's full meaning. Author matching covers lexical, grammatical and rhetorical habits in the requested register; generic polish alone does not satisfy it. Every B task pursues author matching and evasion against Pangram and the common subnet AI-origin panel defined in Section 1. The miner returns exact revised text or an abstention. Every B benchmark requires target-author references, style calibration and the complete detector evidence. Opponent selection and scoring are fixed by the benchmark, independently of the submitting miner. Author-directed revision of one's own or licensed writing and authorized detector-robustness research motivate this task. Passing a detector does not replace the documented production history or any disclosure required for the intended use.

The proposed starter is an 8B decoder, initially Qwen3-8B [27], with a style adapter and a candidate-selection process. Named word and grammar deviations can condition the first version. A learned profile prefix is a later comparison, contingent on better results for unseen authors. Deterministic Rust edit rules can propose small changes; the decoder can propose broader restructuring.

Public A artifacts let B miners run local training queries without a protocol query quota, at their own compute cost. They can use published ensembles and retired A snapshots instead of depending on a peer's availability for every search step. Direct gradients are available only where the architecture permits them; differentiable feedback does not guarantee useful discrete text edits. Training tickets serve authorized rewrite requests from A to B under bounded limits. Benchmark submissions retain the frozen candidate and deadline rules. B can compute its own candidates' A scores before and after commitment; these public-model scores are not hidden feedback. The official comparison uses the one committed candidate and the frozen panel. Hidden labels, other miners' candidates and preservation-review evidence remain restricted.

B submits the complete self-scored probability vectors and declared scalar projections for every assigned A artifact, binding its hash, settings, exact source inputs and final candidate. Validators independently run those artifacts on the same inputs and compare the canonical outputs; they do not reproduce B's private generation process. Every credited B entry additionally supplies the execution receipt specified below, binding its candidate to the registered private model version. A miners need not serve inference calls or help calculate these benchmark scores.

Train first on reviewed source/reference/brief/revision tuples. Then learn preferences among candidates that pass preservation review. Include originals, failed rewrites, copied reference passages, and fluent meaning-breaking edits as controls. Also include polished revisions that erase the writer's voice and detector-passing revisions that retain unwanted filler or formulaic phrasing. The miner's own style score may guide its search, but cannot approve its output for benchmark rewards.

Transformation briefs may specify audience, clarity, accessibility, organization, rhetorical force, and tone. Reducing reading level or length is useful only when the brief calls for it. Suitable text can remain unchanged. Validators use independent judges and reader audits to assess preservation, readability, and requested voice; reviewer ties and disagreements remain in the data.

Future comparisons of 4B, 8B, and 32B transformation models should use fidelity, reader preference, and cost per accepted revision. All checkpoint licenses and revisions must be pinned before a training release. The named starters are proposals, not trained subnet products.

4.3 Private B versions and attested serving

B miners compete through private model weights and paid transformation service. Publish benchmark results and version commitments. Any miner may voluntarily publish its B model; customers access private versions through qualified attested serving. Model secrecy prevents direct downloading; it does not prevent detector providers from buying outputs, studying rewrites or learning similar transformations independently.

Every emission-eligible B entry and paid B offer requires qualified attested execution. Ordinary generation remains available for research and training, with no B emission credit or paid-service qualification. The entry commits to a complete inference manifest before its qualification tasks are disclosed. The manifest identifies private model content by a canonical cryptographic digest over weights, adapters, tokenizer, prompts and other inference data. It separately pins the public runner build, approved architecture and operators, profile construction, search/selection rules, decoding settings and resource limits. The approved loader computes the content digest inside the protected environment and authenticates every loaded page or tensor. It must not merely sign a hash supplied by the miner. The attestation chain binds this loader and its enforced configuration to the instance keys. Customers can thus compare version identity with benchmark records without downloading the model; they cannot independently inspect its private tensors.

Only reviewed, reproducibly built execution code qualifies. Private weights and prompts are treated as data with checked formats and sizes. Miner-supplied executable plugins, arbitrary deserialization, tools, network calls, logging hooks and mutable host mounts cannot enter through a model package. New architectures or search implementations need a separately reviewed runner version. This restricts execution choices in emission-eligible and paid serving; a hardware quote alone cannot make arbitrary private code safe. Independent security review must test malicious model data and inference-engine exploits as well as host attacks.

Benchmark association and serving eligibility. Attestation is an admission gate for B emissions and paid serving. It earns no quality bonus and creates no third mechanism or reward pool. Every credited candidate needs a valid receipt from its frozen private version and qualified runtime, followed by the same author-fit, semantic, Pangram and public-A checks. A change to weights, prompts, search policy or runtime requires a new version and qualification.

The benchmark measures one bounded service invocation. Pin the generation, internal candidate count, local scoring and final selection procedure inside the measured runtime. The task fixes a seed through a versioned derivation from its immutable context. Repeating or restarting that task must reproduce the same result under the qualified execution profile; qualification must demonstrate this property. Miners may train and search freely before freezing an entry, but cannot select the benchmark answer from additional external generation attempts. Required Pangram reports are obtained for that one final text outside the generation runtime. Repeated provider observations retain the declared report-selection policy.

Advertising a benchmark result requires the same model, runtime, search procedure and workload profile in the paid offer. Cheaper configurations, different token budgets or newer private models cannot inherit that score. Publicly identify the tested task population, limits, date, failure rate and strict-success rate separately from incremental utility. This binds a service to measured performance; it does not guarantee the quality of an unseen customer revision.

Miners seeking B emissions must reserve capacity and fulfill assigned training rewrites with that qualified version and serving profile. These real requests test continued service without requiring paid sales or a particular market price. Bind generation receipts on those tickets too. The authorized benchmark/training envelope may return text to its named review recipients; the customer envelope may return text only to the customer. Enforce the signed purpose and recipient outside model inputs. Do not expose task-role or settlement metadata to generation. Recognizing benchmark content remains a qualification risk.

The receipt signs protocol, task and assignment context, instance and model identities, runtime/policy hashes, decoding/search budget, salted input/output commitments and the exact candidate binding. Its signing key is generated inside and bound to the attested instance. The runner signs only output it generated for that accepted input; an API that signs miner-supplied text does not qualify. Validators check the receipt and attestation chain, open the authorized benchmark commitments, rerun A, and conduct the existing independent review. Missing or invalid required receipts invalidate a B submission; ordinary execution cannot earn credit. Receipt-verification disagreement follows the common unresolved-evidence rule. Each valid receipt associates execution with a version under the hardware and software assumptions; it proves neither meaning preservation nor a favorable Pangram result.

The same private version can be provisioned to multiple qualified instances without releasing its weights to customers or validators. The model owner retains its original weights. Model decryption keys may be released only after the owner verifies the approved instance; customer input keys follow the separate customer-controlled release rule in Section 6.2. These are independent authorizations: the owner's approval cannot release a customer's key. The model owner receives no customer input key, plaintext log, retained job state or output copy. Results from a newer private version cannot inherit an older version's benchmark association.

The attested version registry coalesces identical measured B content within the B task pool using the earliest accepted finalized commitment, with canonical UID order for ties. This requires the verified loader's digest and excludes signatures and publisher metadata. Near copies remain an incentive-analysis problem; differing hashes or receipts do not establish independent ownership. Attestation does not solve Sybil registration or the validator-quorum assumption.

5 Training data and evaluation separation

The two launch tasks need three linked data tables, with different supervision:

Table	Records	Supervision
Authorship	Multiple independent works per writer	Identity, date, genre, work family, provenance strength
Production	Original writing, generations, and revision chains	Recorded workflow, parent links, generator version, assistance order
Revision preferences	Sources, briefs, candidate edits, and reviews	Preservation, target voice, editorial preference, ties

Maintain distinct training, public development, private evaluation, and customer reference collections. Split authors before selecting vocabulary or fitting weights. Within evaluation authors, use independent earlier works as references and later works as queries. Keep every revision, translation, and excerpt of one source family in the same split. Hold out generator families and prompt families for origin evaluation. Match content, length, and genre across origins so that a detector cannot win by identifying the assignment format.

B's scored benchmark includes human-only, model-only and mixed sources. The curator freezes the source-origin mixture and assigned aggregation weights before task issue, and validators report results for each origin alongside the aggregate. Every source receives the same author-fit, preservation, readability, target-tone and detector requirements under Section 8.4.

A artifact publication does not require releasing its training corpus or recipe. Qualification uses independently labeled hidden material, with targeted-trigger variants and controls for favoring a paired B. Models receive only declared text and reference inputs, not miner identities or assignment metadata. This removes an explicit cue without making the content blind to its production history or benchmark role.

B training can query published A ensembles and retired snapshots locally at the miner's compute cost. Measure transfer on fresh source and author families and later detector versions. Keep external Pangram observations and query counts, including failed attempts; success against an available A model alone does not establish transfer to Pangram or unseen detectors. Active evaluation material has no general training release; disclosing a source for B generation permits its declared task use. B can compute public A scores on that source and its own candidates. Hidden labels, unreleased sources and other miners' candidates remain restricted until their declared retirement and release.

Mandatory training tickets run in one direction: an assigned A requester supplies authorized source text and a brief, and its assigned B provider returns a constrained rewrite. Retain the source's production record and record B generation as a separate linked revision. Keep source families within their declared splits. B obtains detector feedback by executing published A artifacts locally. Retired benchmark revisions can later supplement A training under their release terms.

Global Voices supplies a prospective pool of attributed modern prose under its default CC BY terms. Qualification must distinguish original writers, translators, shared accounts, and third-party material. PLOS articles can supply factual source material and citation constraints, while their coauthored nature makes them unsuitable as unqualified single-writer labels. Item-level rights and attribution remain part of acquisition. Global Voices policy [28], PLOS article policy [29].

Historical publication alone does not prove a human-only workflow. For stronger labels and useful editing supervision, the proposed pilot commissions 50 writers to provide twelve independent pieces each and two edits of supplied drafts. Common briefs across writers permit comparisons with similar subject matter; multiple subjects per writer test consistency beyond topic. Record collaboration and AI assistance, and obtain separate rights for training, private evaluation, and any redistribution.

If collection and review prove workable, expand to 500 writers: 300 training, 100 development, and 100 private evaluation authors. A further cohort is needed for unknown-author rejection and stronger confirmation. These are acquisition targets; the writers have not been commissioned and this paper commits no funds.

Generate model-only and mixed examples from permitted sources while retaining exact prompts, model revisions, outputs, and human editing records. Synthetic restoration pairs can start from a writer's original, create a generic draft, and train a return toward the original. Review whether that draft still contains the required information; otherwise restoration teaches invention. Keep such weak supervision separate from real writer edits.

Rewrites that fool Pangram or a subnet panel detector are hard examples for later detection rounds. Adversarial benchmark records retain source provenance and assign each candidate a production label from its full revision history. Evasion does not turn model-origin text into a human-only training example. They may become later detector-training examples only after retirement from private evaluation and under agreed release terms. Preserve difficult human examples alongside them. Customer text, profiles, and outputs receive no automatic training or benchmark reuse.

The existing Blog Authorship Corpus experiments remain local noncommercial research and do not supply the commercial starter dataset. Private book and correspondence material is also excluded. Source manifests must distinguish text rights from loader-code licenses. The detailed acquisition plan [30] records the current source screening and outstanding qualification work.

6 Confidential customer inference

6.1 Public A models and private B inference

An ordinary hosted job encrypts text to an operator-controlled miner key. That miner sees plaintext and could leak or exploit an unpublished acquisition announcement, earnings release or funding announcement. Recipient encryption and contractual promises cannot prevent that authorized endpoint from copying the input.

All paid B customer jobs require attested inference: the customer encrypts to an ephemeral key generated inside a verified trusted execution environment (TEE). The operator supplies compute and holds its model weights, but has no protocol key for the customer's plaintext. Both the weights and customer content must remain private under the accepted hardware and software assumptions. Ordinary hosting is available for A when the customer explicitly accepts operator access; it is ineligible for paid B customer jobs or B emissions. Ordinary B execution remains available for research on separately permitted data. Attestation is an execution requirement, with no additional reward pool.

A remains downloadable for local use. Customers may run a voluntarily published B model locally, but the subnet promises no public copy of the strongest private B model. Local inference uses the customer's compute and creates no subnet job or fee. Attested hosting instead depends on qualified hardware, accepted software measurements and the customer's verifier. Its guarantees are conditional on those components.

6.2 Independent model and customer key release

The B model owner and customer authorize different secrets. The owner encrypts its versioned model package before uploading it to storage or a compute host. Its key service verifies fresh evidence for an approved loader, runtime and protected CPU/GPU path, then releases the model key through a channel bound to that instance. The loader decrypts and checks the package inside protected memory and authenticates the loaded version under Section 4.3. The customer's later verification must bind that loaded version to its job key and accepted policy.

Model-key release does not authorize customer-key release. The customer independently verifies the instance before encrypting its input. The model owner's key service has no role in decrypting customer jobs; it receives no customer key or content-bearing job state. The owner can withhold future provisioning but cannot use that authority to change the customer's accepted runtime or recipient. Provision model material before customer secrets arrive and close provisioning interfaces for the job.

The same protected runtime holds both plaintext inputs during computation. The miner may own the hardware or rent confidential compute from another operator. The separate host obtains neither secret; a miner that owns the model still knows its original weights but obtains no customer plaintext. Customers receive encrypted revisions, with no weight download, tensor inspection or debug interface. Keep the execution code public and reproducible while treating private weights and prompts as constrained data. Paid queries can still support behavioral imitation or model extraction; confidentiality does not promise immunity to those attacks.

Encrypted model provisioning with attestation-gated key release has an existing confidential-computing reference architecture [31]. The independent customer channel and restrictions on miner-controlled model data remain requirements for our implementation and security review.

6.3 Attestation and customer key release

The design follows the separation between hardware evidence, verification policy and a relying party's decision to release a secret in the RATS architecture [32]. The customer is the relying party for its input key. Neither a miner's TEE-enabled flag nor a validator's assertion alone authorizes decryption. The following is a proposed application protocol, not an implemented attestation service.

1. The customer selects a signed offer binding miner identity, private model version, accepted runner/policy hashes, hardware class, output key, limits, price and expiry. The client pins its accepted policy and vendor trust roots before processing an untrusted offer; the offer cannot choose its own reference measurements.
2. The selected instance generates a fresh job-scoped encryption key and receipt-signing key inside protected memory. It returns evidence bound to the customer's unpredictable nonce, those keys, the offer and assignment context, the runtime policy and verified model manifest. The customer checks signatures, freshness, endorsements, revocation, security-version floors and disabled debug access before sending text.
3. The CPU evidence must cover the boot chain and enforcement of the complete workload policy. For GPU inference, verify the attached GPU evidence and its protected channel to that same confidential VM. Pairing evidence from an unrelated healthy GPU or ordinary GPU memory does not qualify. Pin the full CPU/GPU/driver/firmware combination.
4. After successful verification, the customer encrypts directly to the attested job key using the bound HPKE context. No gateway, model-owner key broker, validator, fleet-wide recovery key or alternate instance receives a customer decryption capability. The instance accepts only the bound job and verifies the customer's signature over the output key.
5. The measured application decrypts, runs the private model, encrypts the result only to the customer, and signs the receipt binding the input, output and execution version. It then discards job keys and transient state according to the enforced lifecycle. Public settlement sees permitted metadata and salted commitments, not text or raw text hashes.

An initial hardware candidate is a confidential VM using AMD SEV-SNP or Intel TDX with a supported NVIDIA confidential-computing GPU and protected CPU/GPU transfers. NVIDIA documents specific supported configurations; qualification must pin one tested combination rather than infer coverage from a GPU product name [33]. GPU attestation checks device security state and firmware. Application and model binding require the measured loader and execution policy described in Section 4.3; a GPU quote by itself supplies neither. CPU-only enclaves with unprotected GPU offload are insufficient.

Hardware enforcement limits host memory access; customer secrecy cannot depend on the operator declining to inspect RAM. Qualify the exact CPU memory-isolation mechanism and GPU access controls. For example, H100 confidential mode protects GPU memory with hardware firewalls and encrypts CPU/GPU transfers; its on-package HBM is not encrypted [34]. State which physical attacks the accepted platform excludes. An attestation report cannot establish the physical integrity of a hosting facility. AMD documents a physical memory-aliasing attack outside its SEV-SNP threat model [35]; a generic TEE label is insufficient for an unrestricted physical-adversary claim.

6.4 Application isolation and remaining trust

Protect input parsing, tokenization, author-profile construction, generation, search, postprocessing and result encryption inside the accepted boundary. The miner must have no guest administration, debugger, shell, plaintext log, crash dump or command channel into that workload. Disable outgoing tools, external model APIs, Pangram calls and telemetry. Load model material before accepting customer secrets, make model state immutable for the job, and prohibit online training or exporting modified weights. The approved application exposes only authenticated job input and customer-encrypted output interfaces. Attesting malicious application code would preserve its ability to leak.

Use isolated job state, with no prompt/KV cache reuse between customers and no content-dependent public error details. Clear transient buffers before reuse; forbid host-readable swap or snapshots and unsafe migration.

Protected caches can still expose information through timing, so cache isolation is an application requirement beyond memory encryption [11]. Audit private model inputs as untrusted data and restrict their operator graph and execution budget. Malicious weights can influence outputs or timing; encryption of returned text does not prove absence of every covert channel.

The initial private B profile must therefore use a qualified fixed operator graph, fixed tensor-size and token-budget buckets, padding after early completion, and one padded response at the offer's fixed release slot. Keep success and failure envelopes the same public size and expose no token stream or model-selected completion time. This limits deliberate leakage through output length and early stopping. It costs compute and latency; microarchitectural and host scheduling side channels still need independent evaluation. Adaptive graphs or finer-grained metering require a separately qualified policy.

The client verifier and its update path are part of the trusted computing base. Publish its source, reproducible builds, accepted measurements, vendor roots and security-version policy for independent inspection. Certify reference-policy changes under the strict validator quorum and activate them prospectively; the customer must accept the new policy before releasing another key. An operator-controlled web page must not receive plaintext before this verification. Pin the verifier locally or in a customer-controlled application. Treat generated text as untrusted content: escape active markup, disable remote images, link previews and automatic URL/tool fetches, and keep document rendering and profile storage inside the customer's accepted boundary. Malicious revisions could otherwise cause the client to disclose source facts through a network request after successful confidential inference.

Require fresh challenge evidence and current revocation collateral before key release, with fixed maximum evidence age and job duration in the accepted policy. A failed, stale or unavailable verification stops the job before disclosure. Restarts or model/runtime changes invalidate the instance keys and require fresh attestation and customer authorization. Disable resumable customer state in the initial profile; refuse new key releases to revoked versions. A patch or revocation cannot undo a secret already exposed through a vulnerability.

Hardware vendors, firmware, the accepted runtime and verifier, and the customer's own device remain trust dependencies. Timing, size buckets, chosen service, prices and availability remain observable; padding and bounded scheduling reduce some leakage without proving its absence. The design does not claim protection against every side channel, physical attack or future hardware exploit. Any miner can refuse service or withhold ciphertext. Attestation does not prove writing quality, guarantee availability or settle payment alone. B settlement additionally requires complete ciphertext publication and a strict weighted delivery certificate under Section 7.3. Paid B launch depends on an independent security review and the measured acceptance criteria in Section 14.

6.5 Envelope, assignment and recovery

Use HPKE with X25519/HKDF-SHA256, HKDF-SHA256 and ChaCha20Poly1305, plus a separate customer signature; HPKE Base does not authenticate the sender [36]. Bind protocol, chain, subnet, contract, job and assignment generation, offer hash, recipient key, execution mode and policy, model version, output key and expiry. For ordinary A hosting, a hotkey-signed miner key record supplies the recipient binding; that mode grants the operator plaintext access. In attested mode the key binding additionally requires the evidence above. No alternative recipient envelope is supplied to validators or the model owner.

The encrypted payload includes source text, reference works, fitted profiles, brief, private metadata and commitment openings. Use canonical encoding and a fresh secret 32-byte salt for each commitment:

$$C = \mathrm{SHA256}\big(\mathrm{domain} \parallel \mathrm{context} \parallel \mathrm{salt} \parallel \mathrm{length} \parallel \mathrm{payload}\big). \tag{1}$$

Raw hashes, filenames, descriptive titles and report URLs stay inside the encrypted payload. The customer alone receives the result envelope, including probabilities, explanations and edit notes. In attested mode, the measured runner supplies it; the host cannot replace its output key.

A retry on another instance or miner requires a new customer-authorized assignment, successful verification for the selected mode, and a newly encrypted request. There is no automatic plaintext forwarding, key escrow or downgrade. Loss of a job key can make that job unrecoverable; the payment policy handles expiry separately. Customers may provide evidence to a chosen validator under the next subsection, but no audit exception supplies automatic customer access.

Public A artifacts may be evaluated inside the protected instance. Confidential customer jobs cannot send text to Pangram without a separate customer-authorized disclosure outside this protocol. The attested receipt proves no Pangram pass. Authorized subnet benchmarks have their own recipients and external reports; their success is not a per-document guarantee for an unseen customer revision.

6.6 Customer-directed evidence disclosure

Any customer can ask a specific validator to inspect a job by creating a new evidence packet encrypted to that validator's authenticated key. The customer signs a disclosure context binding the job, selected validator and key version, purpose, scope, and validity window. The packet can contain source passages, the delivered result, the miner's signed response, and relevant commitment openings. The customer selects what to send. Neither the miner nor a gateway can activate this access on the customer's behalf.

This grants the selected validator access to that packet, with no decryption capability for other jobs or undisclosed material. The validator must not forward it to other reviewers or providers without further customer direction. Expiry ends authorization for new processing; it cannot erase plaintext the recipient already obtained. Keep the packet, findings, and openings private; publish only any agreed decision or commitment needed by the protocol.

The validator checks which supplied evidence opens the original commitments. A redacted excerpt does not open a commitment to a complete document. Full openings or a separately specified selective-disclosure commitment scheme are needed for stronger binding. Incomplete evidence can support a limited opinion, but cannot establish fidelity of the whole job.

For detection, that validator may reproduce a published A artifact on the supplied material within the customer's disclosure scope. This grants no access to other customer text and does not independently reproduce private B weights. An attested job's receipt may also be checked within the disclosed scope.

Inspection is advisory by default. The initial B escrow settles through execution and encrypted-delivery evidence under Section 7.3. Editorial refunds or revisions require separately accepted terms; refunds are separate transfers and cannot reopen settled escrow. Choosing a reviewer does not redirect principal, extend deadlines or grant automatic disclosure. This route preserves the accepted runtime and customer-controlled input boundary.

7 Paid asynchronous jobs and on-chain coordination

B transformation is the primary commercial product: author-directed cleanup that preserves meaning and target tone while pursuing the required detector objectives. Customers pay for access to competing private models, accepted-revision performance, convenience and qualified attested execution. Miners retain private weights while offering newly benchmarked versions. A remains an open detector resource and a secondary hosted service for customers who prefer an API to running it themselves. The market determines whether A hosting is cheaper; the protocol mandates no price ratio.

Miners earn network emissions for independently evaluated benchmark utility and may separately earn customer fees. These are different revenue sources. Customers can pay for B without receiving its weights; benchmark version commitments and attested receipts support that service claim. Hardware attestation supplies no exclusivity, guaranteed premium or permanent advantage against detector providers studying served outputs. Reader outcomes, repeat purchases and margins after confidential-compute costs must test the business thesis.

The proposed subnet settlement transfers the quoted fee to the serving miner, subject to its disclosed fees and payment rules. It does not automatically pay a royalty to Slop Ninja Research or make customer revenue an emission multiplier. Any future application subscription or gateway fee needs explicit customer terms; none is assumed in the protocol's economics. Native subnet-owner emissions are a separate allocation under chain policy [37]. They belong to the recognized owner and any applicable lease arrangements, not automatically to Slop Ninja Research or an investor. A supplied subnet slot therefore needs explicit ownership, funding and revenue terms. Alpha-denominated demand alone establishes no token-price or investor-return guarantee.

The commercial pilot must separate recurring inference margin from research subsidies. For B, record customer fees less serving, protected capacity, delivery and transaction costs; separately record training, mandatory rewrite tickets, benchmark generation and Pangram costs against emissions or named research funding. For validators, measure artifact execution, ciphertext handling, report retrieval and full semantic review against their own funding and expected receipts. Fixed job quotes must cover reserved resources, including padding and failed work. Neither claimed GPU expenditure nor paid volume increases benchmark rewards. A free-request quota is viable only if its bounded cost supports continued participation; measure it before freezing the launch workload.

Offers identify execution mode, model/version evidence and the accepted runtime policy. All paid B jobs require attestation under Section 6.3. Ordinary B customer offers are ineligible even when an operator describes the text as non-sensitive. A customers may explicitly choose ordinary hosting and grant the operator plaintext access. Public A and voluntarily published B models can run locally, without a subnet job or fee. No public copy of the best private B model is promised.

B jobs pursue author matching and detector evasion within the selected execution boundary. Such a job has no verified Pangram result unless the customer separately authorizes external disclosure; its objectives do not expand recipient access. Authorized benchmark rounds provide the routine external measurements under Section 11.1. The proposed job contract uses Subtensor's EVM to coordinate offers, escrow, commitments, and settlement. Inference prices are denominated in Slop Ninja's subnet alpha; execution stays off-chain. Subtensor EVM guide [38].

7.1 Market pricing in alpha

Miners publish signed offers for bounded jobs. Each offer specifies a total alpha price, task, service version and execution mode, input/output limits, delivery deadline, available capacity, quote expiry, and settlement terms. Attested offers also bind the padded workload bucket and fixed response-release slot; charge the reserved job price without publishing content-dependent token usage. Customers select among compatible offers using price, benchmark performance, and delivery time. A higher-quality model or faster service may command a different price. There need not be one clearing price for every kind of inference.

Miners may revise their unreserved offers as demand, queue length, compute costs, and competition change. The subnet operator sets no base price, utilization target, or global price-adjustment coefficient. Miners choose their own pricing policies; customers decide whether the offers are worth buying. Alpha is the unit of account and proposed settlement asset. The inference market determines the number of alpha units charged independently of the alpha/TAO exchange market.

An atomic reservation consumes offered capacity and locks the quoted alpha amount and job terms. Later price changes apply only to later reservations. An expired or exhausted offer cannot be filled; repricing requires a new customer choice. The reservation therefore gives both parties a known price through execution and settlement, even while other offers change.

Quote discovery uses public service descriptions and the size/deadline information the customer chooses to reveal. Source text, reference writing, and the full brief remain encrypted to the selected instance or, for explicitly ordinary A hosting, its operator-controlled key. Obtaining competing offers does not distribute the plaintext to prospective bidders. If the selected miner cannot serve the bounded request, the job follows its rejection/refund terms; it cannot silently raise the price after decrypting the input.

7.2 Reservation and delivery

Subtensor's staking-v2 interface exposes allowances and stake transfers, and contracts can hold stake through their mapped coldkeys. The proposed escrow adapter would use those operations to custody unlocked, transferable alpha on the same subnet. An allowance alone does not fund a reservation. Alpha remains associated with a subnet and hotkey; settlement must account for that position and its native 10^{-9} precision. These primitives do not supply a finished job-escrow implementation [39,40].

An offer distinguishes the service price from network and transfer fees and names the fee payer. Funding succeeds only after the credited alpha principal satisfies the quote. Release and refund must reconcile actual balances, transfer permissions, and fees; incidental stake accrual must remain separate from job principal. The adapter and its accounting policy require validation before paid launch. Neither an alpha price feed nor an assumed ERC-20 interface substitutes for that work.

The B lifecycle is:

1. **Verify and reserve.** The customer verifies fresh instance evidence, the registered model/profile and the job encryption and receipt-signing keys. Its signed reservation pins those keys, the accepted security policy, validator snapshot, terms and quoted alpha principal.
2. **Submit and accept.** The customer delivers the input envelope and salted commitment. The protected runtime validates the opening, schema, limits and customer-authorized output key, then signs acceptance. The host relays that acceptance to the contract by the fixed cutoff.
3. **Execute and publish.** The approved runtime runs the agreed computation, encrypts the result to the customer and signs its completion receipt. The host publishes that receipt and the complete padded ciphertext through the job contract within the agreed release window.
4. **Certify and settle.** Validators verify execution evidence and encrypted delivery. A timely strict-weight certificate releases payment under the checks below. Customer acknowledgment is unnecessary. Failure to meet the fixed acceptance, delivery or certification cutoff permits the specified refund.

Bind every signature and state transition to protocol version, chain, subnet, contract, job ID, participant roles, nonce and assignment generation. Authenticate the hotkey-to-settlement-identity association; a copied or truncated address is insufficient. Pin the canonical receipt encoding, signature suite and contract verification test vectors. Subtensor exposes signature-verification primitives, not a completed TEE verifier or job escrow [41].

Input may travel through authenticated encrypted transport; validators receive only its salted commitment and the runtime's acceptance binding. For the initial bounded B pilot, a single completion transaction supplies the entire customer ciphertext and receipt. The contract checks its length and hash against the receipt, records completion and retains the retrievable bytes in transaction data or events. A URL, hash, partial upload or enclave signature alone cannot establish delivery. Invalid publication must not consume the valid completion slot. Any party may relay the exact signed completion without changing its recipient or payee.

Measure 4, 16 and 64 KiB padded result envelopes, receipt overhead, gas, finalized-history retrieval and client recovery before freezing the maximum job size. Admit only jobs that fit the qualified bound. On-chain ciphertext exposes metadata and remains vulnerable to later key compromise. Off-chain availability requires a separately specified and qualified protocol; it is not an equivalent pilot fallback.

7.3 Receipt-certified B payment

B payment purchases the agreed bounded computation and retrievable customer-encrypted result. Validators certify these facts without receiving source text, result plaintext, private weights or commitment openings. Customer satisfaction is not a default payment predicate. The attestation verifier evaluates evidence against the accepted policy; a hardware signature alone does not approve the application [32].

Execution evidence. The protected runtime signs acceptance only after checking the submitted input and output key. Its completion receipt binds that acceptance, the input commitment, job and offer, miner and settlement payee, model content digest, runtime/policy and workload profile, instance evidence, customer output key, padded ciphertext length and hash, and deadlines. Only a successful run of the agreed procedure may produce a completion receipt. The runtime must generate and encrypt the result itself; it cannot attest arbitrary miner-supplied text, truncated execution or a model-selected error as completed work. The model's poor revision can still be the valid output of that procedure.

Customer authorization fixes both instance keys before disclosure. Validators check their binding to the approved loader and complete CPU/GPU path, the measured model, freshness, disabled debug access and the job's accepted revocation collateral. The reservation pins an authenticated security-policy/revocation snapshot, maximum evidence age and bounded job duration, all still valid at acceptance. Later updates stop new acceptances and do not retroactively rewrite the settlement test for accepted jobs. A newly discovered vulnerability remains a security incident; payment cannot repair an earlier exposure. Restart, key substitution or a changed model/profile cannot silently replace an accepted job.

Validator authority and certificate. Before reservation, derive the complete paid-certifier roster from the epoch's protocol-defined validator eligibility and a proof-verified finalized native snapshot. Do not select a subset by responses to this job or customer/miner preference. Bind each identity, registration generation, signing key and integer effective consensus stake, with total $W > 0$, using the snapshot authentication rules in

Section 10. Exclude the serving miner's exact UID and any customer UID linked by authenticated chain identity. Different keys do not establish independent owners; assume dishonest weight is strictly below $W/3$ in this conditional set.

The paid roster and signing keys serve public receipt checks. They grant no benchmark-style customer envelopes or decryption access. Reserve verification capacity and identify its funder before accepting jobs; a relayer or a small preparer group cannot choose a payment quorum. Missing, abstaining, recused and offline validators remain in W. Every signer independently checks the instance evidence, acceptance and completion receipts, exact offer bindings, and the complete ciphertext in finalized chain publication.

A `DELIVERED` certificate requires distinct eligible signers with agreeing weight w satisfying $3w > 2W$. It binds the job, offer, snapshot, accepted policy, receipt digest, completion transaction and ciphertext hash/length. The contract verifies the frozen signer weights and signatures, the pinned runtime receipt signature and job bindings, matching recorded ciphertext publication and all deadlines. Validator certificates cannot waive those contract checks. No new receipt, output or validator set may replace an accepted completion after issue. Retries of identical evidence are idempotent.

Deadlines and failure. Each funded job fixes acceptance height H_a, delivery cutoff H_d and certificate cutoff H_c, with $H_a < H_d < H_c$, a fixed padded response-release slot and a bounded publication window. The qualified policy supplies minimum execution, inclusion and verification windows; missing capacity or incompatible deadlines prevent reservation. The runtime releases its padded envelope at the fixed slot; late host relay does not change its hash or extend H_d.

Successful canonical transaction inclusion at a cutoff is timely. Validators wait for finalized acceptance and completion evidence before certifying. A certificate must itself be included by H_c; the decision becomes terminal when that inclusion is final. No timely acceptance permits refund after H_a; no timely valid completion permits refund after H_d; timely completion without a certificate permits refund after H_c. Refund decisions wait for finality of the relevant cutoff. A late certificate cannot recover refunded principal. The contract records exactly one payment or refund; retries cannot pay twice, and neither replacement evidence nor a dispute extends deadlines. Miners, customers or keepers submit settlement transactions. Relaying evidence does not change the beneficiary.

With honest, available quorum and timely chain inclusion, a customer cannot consume a result and veto payment by withholding acknowledgment. Validator outages, censorship or insufficient agreeing weight can still cause a refund after valid delivery. The miner bears that bounded risk; retain a `CERTIFICATION_TIMEOUT` record distinct from proven execution failure. No paid-job timeout automatically becomes a mandatory service failure or changes emission credit. Finality stalls can delay terminal settlement; they cannot justify drawing new judges or shrinking W. These dependencies prevent an unconditional fair-exchange claim.

Editorial disputes and A hosting. Completion certificates establish no semantic fidelity, author-fit success or Pangram pass. Benchmark evaluation remains independent. Any customer may separately send evidence to one chosen validator under Section 6.6. Inspection is advisory unless both parties accepted specific refund or revision terms beforehand. The initial B escrow uses the receipt/certificate rule; additional editorial refunds are separate transfers under those terms and do not reopen settled escrow or authorize automatic plaintext disclosure.

Ordinary A hosting cannot supply a protected generation receipt. Its explicitly separate pilot mode retains customer acknowledgment and bounded timeout refunds, with disclosed miner nonpayment risk. A hosted in a qualified attested runtime may use the B delivery-settlement procedure for its declared A computation. A artifact publication and emission eligibility do not require this hosting mode.

7.4 Revenue and costs

Customer fees pay for the purchased service. Emissions reward performance on separate benchmark assignments. Customer volume does not directly increase emission rewards, because miners can transact with themselves. Paid successes cannot dilute failures on mandatory protocol service: the measured availability gate applies in aggregate and separately to each committed interface and assigned role's mandatory work. Its zero-credit and recovery-deficit rules are specified in Section 8.2. The zero-credit effect must be validated under the chain's payout lag; it cannot reclaim already distributed emissions. A fixed quote charges for the reserved workload without publishing content-dependent internal token counts.

Miners pay their own training, inference, private search, and required candidate Pangram submissions for benchmarks. The benchmark operator or participating validators fund source-baseline reports from an explicit round budget; those reports can be reused across competing miners under the same task/version. Validators also bear A artifact distribution and benchmark execution, retrieval, hosting, and review costs. Subsidies, if used, must identify their funder and cap before assignments begin. This design assumes no free API-credit allocation or automatic on-chain reimbursement scheme.

Assigned B rewrite requests consume reserved provider capacity under the training-service obligation in Section 8.2; the recipient pays no per-call fee within that budget. Providers bear the compute cost as a condition of emission eligibility. A miners have requester obligations for those tickets, with no mandatory A inference endpoint. Authoritative A benchmark execution uses the validators' separately reserved budget. B miners may query public A artifacts without a protocol quota during training, bearing that compute cost themselves. Official B generation uses its frozen bounded attested profile. This obligation covers neither Pangram's external charges nor unlimited hosted search. Bootstrap reference artifacts and execution need a separately named research funder. Confidential customer jobs cannot automatically forward inputs or outputs to Pangram or a peer miner.

8 Benchmark scoring and mechanism mapping

The initial subnet has two logical tasks and two chain mechanisms: A, author/origin detection, maps to mechanism 0; B, author-directed cleanup and transformation with detector evasion, maps to mechanism 1. Writing quality and meaning preservation are requirements within B, not a separate launch reward pool. Deployment must respect the documented joint UID/mechanism capacity constraint [37].

For an initial offline tournament, give A and B equal budget shares. Within A, balance author and origin tasks equally. Every B comparison jointly evaluates author-style matching and evasion of Pangram and the qualified subnet detector panel. These are pilot allocations, not an economically optimized split. Publish any live allocation and change policy before the scoring epoch. Isolated style or detector ablations remain diagnostics and earn no B emissions.

8.1 Detection

Validators obtain probabilities by executing the published, qualified A artifact frozen for the round. Its manifest binds the complete inference procedure and reference runtime; endpoint replies cannot replace these executions. Independently labeled hidden tasks determine author/origin performance. Publication establishes which computation is being scored, not the correctness of its predictions [42].

For a valid probability vector p and known one-hot label y, use the multiclass Brier utility:

$$B(p, y) = 1 - \frac{1}{2} \sum_k (p_k - y_k)^2. \tag{2}$$

Average query results within source groups and authors before the declared task mixture. Compare the aggregate with a fixed public baseline, then use positive improvement divided by that baseline's remaining headroom. If the baseline is perfect, the pool has no remaining improvement and is unallocated in the offline simulation. Clip after aggregate comparison, not separately for each lucky prediction. Raw Brier utility is unsuitable for direct payout normalization: a uniform prediction in a large author gallery can already have a high value.

Report retrieval accuracy, reciprocal rank, origin calibration, and human false-positive rates separately. Fix operating thresholds on development data. Unknown or weak production records cannot become hard labels merely because Pangram agrees with a miner.

The quality anchor for A's origin task is selected by detection sensitivity at fixed human false-positive operating points under Section 11.1. The B panel's lead detector is selected separately by held-out origin Brier performance; the two roles can belong to different detectors. Pangram is the initial quality anchor; a superior subnet detector takes over, and an improved Pangram can regain that role. Validators compare A artifacts and Pangram on the same independently labeled hidden origin set and publish performance gaps and uncertainty regardless of who anchors the comparison. Documented production histories determine reward labels; agreement with

Pangram earns no separate utility. The fixed public probabilistic baseline above remains the reward normalization reference. Pangram's content fraction cannot directly replace that probability vector or supply author labels. Selecting this quality target does not change the B panel's origin-Brier ranking or an issued round's artifacts.

8.2 The adversarial feedback loop

Participation includes public A artifacts and a bounded training-service obligation. A miners request authorized training rewrites; assigned B miners provide revisions that follow the source and brief. The request is free at the point of use, with the provider bearing its cost in exchange for eligibility for task emissions. Every credited B provider uses its qualified attested service version for these assigned rewrites, with a bound execution receipt. Expected emissions need not cover every miner's costs. The obligation concerns delivered service, not unverifiable training effort. Independent preservation, readability, and requested-tone review are required for B. No miner can approve its own revision for rewards.

Public A artifacts let B miners train and search locally. Those queries consume B's compute and have no protocol per-query quota. This local access does not create additional benchmark submissions or discharge an assigned service obligation. Validators separately reserve their own compute to grade the official comparison matrix. A earns quality credit from independent evaluation of its published artifact; B earns it from the committed revision. Section 4.1 defines canonical A execution. Every emission-eligible B entry requires the version-bound execution receipt in Section 4.3; validators check that evidence, not private weights. Attestation supplies no additional utility or pool allocation.

Use released, authorized training sources and retain the source history, B execution record, brief, and preservation status with each response. Assign each response a production label from its full revision history. A later rewrite does not acquire a human-only label by defeating a detector. Unknown assistance histories remain weak labels; a private endpoint is not itself proof of model generation. The source-family splits, release rules and access boundaries in Section 5 govern this exchange. Private B weights can still be studied through served outputs; request quotas limit that exposure without preventing model extraction or retention by recipients.

8.2.1 Service budgets and deterministic pairing

Freeze a global budget, request schedule, interface qualifications, and capacity limits before each evaluation epoch. A artifact execution and B rewrite service have separate resource limits and capacity reservations. New UIDs and traffic do not increase the global budget, and serving volume earns no reward by itself. There are no counterparty nominations or voluntary pairing offers. Customer market quotes remain separate from this protocol-assigned work.

For each protocol-issued interaction, consume the next lifetime index i_u of the requester UID slot u. The counter is persistent for that chain and subnet, including through hotkey changes and UID reuse; each assignment also pins the current registration generation, hotkey, service version, and encryption key. A requester cannot skip indices, reset them, or issue empty calls merely to advance the counter. An expired assigned slot still consumes its index and counts as a requester failure if the requester abandons its obligation; retransmission keeps the same index, request, and peer assignment. New or replaced registrations must qualify before earning rewards; inheriting a slot counter does not inherit its former holder's performance record.

Let R_e commit the eligible, active service roster fixed for epoch e. For an A requester, rank each eligible B provider $v \neq u$ by

$$h_{u,i,v} = \text{SHA256}(\text{encode}(D, \text{chain}, \text{netuid}, u, i_u, v))\,, \tag{3}$$

where D is a versioned domain and encode is a canonical, length-delimited encoding. Break hash ties by UID. The first k distinct entries form the small permitted peer set. The first peer is assigned; later entries are prescribed fallbacks following recorded failure, not a menu from which the requester picks a favorable result. The pilot must publish k, capacities, and fallback limits before work begins. If the permitted set is exhausted, retain the failure; do not hash again. Hash-derived ranking and the fixed global schedule must respect provider capacity without allowing discretionary substitutions. Process capacity reservations in a canonical slot order; a request is issued only when its prescribed service windows fit the reserved capacity. Fallback stages have predeclared deadlines and minimum response windows, so later providers cannot inherit an impossible deadline. The ticket pins R_e for eligibility without mixing it into each peer's rank: changing one roster member does not reshuffle the relative order of all remaining peers.

Each ticket binds the epoch, UID/index, roster root, prescribed peers and roles, permitted task, salted input commitment, size limits, deadline, and nonce. Anyone can check the scheduling metadata and sequential consumption against the finalized ledger. The text and commitment openings remain restricted to authorized recipients. Counterparty UID visibility does not establish the origin of the text. Validate authorization before inference; duplicates return the recorded response without another charge against the quota. Counter issuance and quota consumption require one authoritative, replay-protected ledger shared by validators, not independent per-validator counters.

This schedule is predictable. Any miner can inspect future assignments, and another UID may have the same owner. Registration costs, bounded work schedules, and penalties for assigned failures constrain abuse; the hash does not prove operator independence or eliminate strategic abstention. Block height defines protocol-assigned slots and deadlines. Using a requester-chosen submission block as the peer seed would permit waiting for a favorable set. A short deadline narrows that opportunity without making a public block number unpredictable.

8.2.2 Availability from actual requests

The default Bittensor tempo is 360 blocks, approximately 72 minutes at 12 seconds per block; the actual subnet tempo and finalized epoch boundaries govern the schedule [43]. No heartbeat is required. Use authenticated outcomes of real assigned inference requests to measure availability.

The mandatory classes are A as requester of training rewrites and B as their provider. A artifact availability and performance are separate qualification requirements. The schedule must reserve real work for every activated mandatory class before that class can earn availability credit.

For miner u and measured epoch e, let $N_{u,e} = S_{u,e} + F_{u,e}$ count settled successes and attributable failures under Section 10. Retain unresolved obligations separately as $U_{u,e}$; they block eligibility without inventing a failure. If validator non-certification leaves a service obligation unresolved, even an honest miner loses earned emission eligibility for that measured epoch. If this affects every miner, no positive eligible skill remains and the pool follows the unallocated/burn policy below. An obligation is either submitting a scheduled request or answering a valid delivered request; the ticket fixes the responsible UID and role. The emission-eligibility gate is

$$L_{u,e} = \begin{cases} 1, & U_{u,e} = 0,\ N_{u,e} > 0,\ 10F_{u,e} \le N_{u,e}, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

More than 10% failures means zero earned emission credit in both A and B for that measured epoch. Exactly 10% passes this availability gate, while quality requirements still apply. With no assigned requests there is no availability evidence and no automatic pass. Report counts and failure rates by interface as well as in aggregate. Apply the same gate separately to mandatory protocol service for each committed interface and assigned role, and require every applicable gate to pass. Purchased traffic cannot dilute failures on mandatory work. Successful outgoing requests cannot dilute missed provider replies. Each miner needs a positive assigned workload on each interface for which it seeks eligibility; apply a role-specific gate when the schedule assigns that role work. An answer after its deadline remains a recorded late answer. A successful fallback does not erase the preceding provider's decline or failure.

Only protocol-issued requests within the fixed input, output, capacity, and response-time limits enter N. Unsolicited traffic, duplicate retries, malformed inputs, and deadlines that cannot meet the declared service window do not count. Count a protocol-issued requester obligation even if the requester withholds its payload to avoid an unfavorable peer; do not count that undelivered payload as the provider's failure. Bind the request's measured epoch when it is issued; every counted deadline must fall within that epoch's closed measurement window. Requests due later belong to a later window and cannot be marked late early. Section 10 fixes complete ciphertext publication, validity certificates, fallback activation, and incident closure. A counterparty's accusation alone cannot establish a failure. Publish unresolved and void counts alongside settled work; neither retires recovery deficits.

Derive the next epoch's normal active roster from registration, qualification, and the completed prior window's request records. A small, deterministic probation allocation of real requests lets new or recovering UIDs establish availability without a self-declared online flag. Missing evidence can block normal admission without proving misconduct. Freeze each issued interaction's roster and ordered peers; mid-epoch declines or outages invoke its existing fallback list rather than recomputing the set. The probation budget and minimum workload coverage must be measured in the pilot and fixed before live use.

The zero-credit cutoff can remove the immediate incentive to finish an already failed epoch. For each mandatory interface and role, retain a recovery deficit alongside the epoch gate:

$$D_e = \max\{0, D_{e-1} + 10F_e - N_e\}, \qquad D_0 = 0. \tag{5}$$

Only authenticated, protocol-assigned work in that same class enters this update. Normal admission and emission eligibility require zero remaining deficit as well as the current availability and quality gates. Past surplus service cannot be banked; an epoch with no work cannot erase a deficit. Probation uses bounded real requests to permit recovery. Purchased jobs and other interfaces cannot retire the deficit, and key or service-version changes within a registration do not reset it. Dropping an advertised interface or role does not cancel its accrued deficit; every outstanding class must be cleared before normal admission. Replacement registrations follow their own qualification and probation; the design cannot establish whether their owners are different.

For 100 fixed obligations, 11 failures followed by 89 successful replies leave a deficit of 10. Abandoning all 100 leaves 900. Both cases earn no credit for that epoch, but require respectively 10 or 900 subsequent successful obligations to clear the deficit if no further failures occur. Capacity limits determine how many epochs recovery takes. This rule gives continued service a future eligibility benefit; it does not prove that every miner values future emissions above its costs.

The measurement window must close early enough for the relevant weight submission and commit-reveal deadlines. Publish which payout cycle settles each measured epoch. The proposed zero-credit rule cannot retroactively erase emissions already distributed by the chain. The adapter must demonstrate its effect on both mechanisms under actual weight-reveal delays and stale-weight behavior before launch; this paper assumes no custom slashing primitive.

8.2.3 Common evidence for transformation rewards

Per-miner peer sets allocate authorized B rewrite service for A training. Those training outputs do not replace the common committed benchmark or let a requester nominate scoring inputs. The benchmark scheduler reserves validator execution capacity for a common comparison batch using published A artifacts, plus verification of the required attested B receipts. This matrix consumes no miner service-ticket quota or lifetime interaction index.

For each shared comparison index, the benchmark scheduler deterministically selects three qualified A artifacts on distinct UIDs and a B batch from the frozen registry. Admit only qualified attested B entries and freeze each serving profile. Coalesce identical verified B content under Section 4.3 before ranking its canonical UID. Include the highest-ranked qualified origin detector from the previous completed A round in the A panel; hash-rank eligible remaining A UIDs under a separate comparison domain to select the other two and the reserve order. Then hash-rank eligible B UIDs outside that panel to fill the published batch size. Filter the deterministic A reserve ranking against every UID in the B batch and the initial panel, then freeze the remaining reserve order and capacity. Distinct artifacts must have distinct inference-content hashes, with duplicate model-credit entries coalesced under Section 4. A reserve cannot later score its own B entry. Insufficient reserved capacity prevents issuing that comparison; it does not permit another B draw. Qualifications, sizes, capacity, tie-breaking, and the shared counter are frozen and reproducible. Miners cannot nominate a panel, join a preferred batch, or obtain another draw by declining work. Only exact UID exclusion is asserted; different UIDs may share ownership.

The panel's lead detector is the strongest qualified A service by held-out origin Brier performance, subject to the declared calibration and human false-positive limits. Validators freeze that ranking from the previous completed A round. Author-retrieval rank does not qualify an origin service. Every B entry faces the same three detectors, source texts, briefs, authorized target-author references, style artifact and calibration, Pangram policy, source-baseline evidence, and frozen settings. Every B benchmark requires these inputs. Validators validate and cache the entire selected A panel before revealing the B task or starting its generation window. Every B entry must have qualified its runtime, model manifest and complete bounded generation/selection profile before that issue point. Bind its one final candidate to that invocation; external generation retries cannot select a better benchmark answer. Artifact reserves may be used only before that issue point. Validators execute a complete source and candidate matrix after B candidate commitments, using the same frozen artifacts and reference runtime. This is what matched assignments mean. The three-member pilot is provisional; its cost, coverage, and susceptibility to colluding UIDs require measurement before live rewards.

Apply the frozen-artifact and runner-recovery rules in Section 4.1. Unresolved execution follows the candidate-specific or shared-evidence closure in Section 9. Any B miner that fails to submit receives zero on its assigned work. A missing strongest-detector result prevents a claim of success against that detector. Deliberately triggered execution errors are a required pilot attack case.

The median panel improvement used below limits one arbitrary outlier when fewer than half the panel UIDs collude. It does not establish that bound or prove owner independence. Validators reproduce model behavior; the artifact qualification controls in Section 4.1 must test deliberate bias. One A result cannot alone determine B's utility, and A earns quality from independent labeled evaluation, not from how harshly it scores B. The strongest detector's result is also reported separately. Legitimate detector disagreement alone is neither fraud nor a replacement trigger.

Validators execute A under an isolated reference interface that excludes B UIDs, source/candidate role labels, scheduling metadata and undeclared state. Mix human and model controls. Text itself can still reveal its origin or purpose; interface restrictions do not prove blindness. Retain validator-signed execution records binding the model, runtime, inputs and outputs. Miner-selected local search records cannot replace this matrix. Verify attested B receipts against the committed task, candidate, model version and approved runtime. No raw private weights are disclosed to validators. B submits self-scores with its final candidate, binding the assigned panel hashes, reference settings, full candidate probability vectors, required source baselines and scalar projections. These numbers record the miner's claimed result under the assigned configuration, so validators can identify and investigate discrepancies tied to that submission. They earn no separate credit and do not reduce required validator execution. Validators recompute the canonical outputs and compare the claims; verified results determine rewards, with mismatches retained. A claimed high score cannot skip another candidate's verification. B miners commit submission evidence, and validators commit their verification records, by the fixed evidence deadline. Hash-assigned preparers organize each submission's dossier. Every certifying validator independently performs all required checks before signing; final certificates require the strict quorum of the full frozen eligible set. At launch, this process covers every scored submission. Bootstrap rounds use published, explicitly funded bootstrap A artifacts until enough competitive models qualify, and are reported separately.

8.3 Preservation before revision rewards

Each editing task has a preservation brief and hidden review material covering claims, participants, quantities, negation, uncertainty, citations, attribution, and argument relationships. Review the entire revision. Positive utility requires preservation of the source's full meaning and conformity to the task's target tonal intent. That target may differ from source tone when the requested authorial style calls for a change. Validators also check readability and tone under the fixed rubric in Section 9. Before issue, the brief identifies required cleanup of unwanted filler, repetition or formulaic phrasing under R, and intentional voice features to preserve under T, including any specified informality or idiosyncrasies. Cleanup must retain the protected information. These are B acceptance constraints. A certified failure sets $G = 0$; certificates for all required passes set $G = 1$. Every task requires tone review, including a request that supplies no separate tone instruction. An unrequested tone reversal can defeat the writing brief while retaining literal claims, so we require this vote and measure its added review cost and void exposure in the pilot.

Readers compare anonymized revisions in hash-derived order without detector or miner scores. Maintain unchanged sources and qualified editor baselines. Automated judges join reward allocation only after agreement and failure modes have been measured against separate reader judgments. Instructions embedded in a candidate are evaluated text, not evaluator instructions.

Unresolved candidate judgments at the fixed close receive zero credit for their assigned slots under Section 9, while competitors with complete evidence retain their scores. Miner nonresponse also remains zero on assigned work. A certified shared failure affects the full declared scope. Keep assigned aggregation weights and publish coverage so conditional accuracy cannot conceal difficult cases. Unresolved mandatory-service validity is a separate matter and can still block epoch eligibility. Selective non-certification and honest semantic disagreement remain review risks.

8.4 Transformation utility

Human-only, model-only and mixed sources are all eligible for scored B comparisons and emission credit under this utility. Source origin grants no exemption from either objective or any acceptance gate.

Let $V \in [0,1]$ be the exact normalized positive improvement in target-style fit defined in Section 9. The frozen public evaluator returns integer style-fit scores q_x for source x and q_z for candidate z, using the same target-author references; private miner predictions cannot change them. Validators validate the evaluator against independent reader comparisons before live use. Topic copying, reference copying, and meaning-breaking edits must fail that validation. No miner-defined embedding can be its own reward oracle.

For Pangram evasion, let $F_r(t)$ be the reported AI-plus-assisted fraction in required observation r for text t. The initial benchmark requires three distinct completed reports for each source x and candidate z. Improvement using the least favorable pair among those submitted observations is:

$$\Delta_P(x,z) = \min_r F_r(x) - \max_r F_r(z). \tag{6}$$

The strict target additionally requires $\max_r F_r(z) < 0.10$. Three reads of one report count as one observation. The 10% cutoff is the project's declared engineering target. Any tolerance used to compare repeated reports must be reported separately and cannot widen that success criterion. Use the declared provider version and repeat policy for both sides; report sensitivity to provider variation.

The qualified runtime produces one final candidate under the bound invocation profile. Miners may pay for additional Pangram report attempts on that exact text before commitment. Scoring uses the committed reports, including variation in their provider results. The protocol does not require disclosure of every provider call or a fresh validator-paid Pangram inference call. A reported pass applies only to the committed observations, not to every future measurement of that text. Selection can bias the submitted set when provider results vary. We accept this limitation to keep external Pangram costs with the submitting miner. Validators independently execute the frozen A panel against the committed candidate. Our probe found equal document fractions across fresh requests for three tested texts, with auxiliary-score variation, but no observations within two percentage points of 10%. It establishes neither general determinism nor threshold stability [44].

For panel detector j, let $H_j(t) = p_j(\text{model-only} \mid t) + p_j(\text{mixed} \mid t)$, using its validator-verified probability vector under the frozen policy. These are document-origin probabilities; Pangram's F_r is a content fraction. Define $\Delta_j(x,z) = H_j(x) - H_j(z)$ and $\Delta_S(x,z) = \text{median}_{j=1}^{3} \Delta_j(x,z)$. Retain each detector's raw responses and improvements. The pilot evasion utility is

$$E(x,z) = \begin{cases} (\Delta_P(x,z) + \Delta_S(x,z))/2, & \Delta_P(x,z) \geq 0 \text{ and } \Delta_S(x,z) \geq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

The strict detector pass additionally requires $G = 1$, $\max_r F_r(z) < 0.10$, a majority of panel detectors satisfying $H_j(z) < \tau_j$, and a pass against the strongest qualified prior-round origin detector at its own threshold. Calibrate each τ_j on independent development data at the declared human false-positive operating point and freeze it before the round. These thresholds are not automatically 0.10. Report strongest-A and panel pass rates separately; the strongest-A result is not a sole veto on incremental utility. Neither Pangram improvement nor panel improvement compensates for regression on the other aggregate axis.

When the required evidence is resolved, the sole B utility is

$$U_B(x,z) = \begin{cases} G(V+E)/2, & q_z \geq q_x,\ \Delta_P \geq 0,\ \Delta_S \geq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Check these raw conditions before awarding credit: clipping a negative improvement to zero in V or E cannot conceal a regression and allow the other component to earn a reward. Both author fit and detector evasion are mandatory parts of every comparison. The semantic gate requires preservation of full meaning, readability and target tonal intent; neither component can compensate for a failed field. Unresolved evidence follows the common closure rule. Any declared unscorable candidate earns zero B utility under Section 9.

Incremental utility can reward improvement in one component while the others remain unchanged. It does not certify an absolute author-fit threshold or the strict detector target. Report author fit and detector passes separately from incremental utility; a strict pass must also meet the raw nonregression conditions above. An already suitable source can stay unchanged without receiving an invented improvement bonus.

If any required source report gives $F_r(x) = 0$, Pangram nonregression requires every candidate report to give $F_r(z) = 0$. This applies to human-only sources as well as model-only and mixed sources. Style improvement may still earn incremental utility when all gates pass, including $\Delta_S \geq 0$. A human-only production label neither substitutes for measured detector evidence nor earns credit by itself.

Validate this proposed utility against blinded reader comparisons before live reward use. It must order useful revisions above omissions, unsupported strengthening, irrelevant padding, and needless edits.

8.5 Aggregation and failure policy

Fix assignments, task mixtures, baselines, and deadlines before submissions. Balance related examples within source groups using their preassigned weights. Failed, missing, unresolved and scoped-void slots contribute zero without disappearing from the denominator; publish their distinct statuses. Extra cheap submissions do not increase a miner's share. Apply the availability and zero-deficit gates to each miner's earned A and B skill for the measured epoch. B entries additionally require valid qualified-version receipts and fulfillment of their reserved attested service obligations. Ordinary B entries earn no emissions. Then normalize eligible positive skill within each task pool and combine using the published shares. The chain mechanism split carries the corresponding aggregate task allocation.

If no miner beats a baseline, record that pool as unallocated in the offline simulation. The proposed live fallback directs that pool's weight to a verified owner-associated hotkey whose miner incentive is withheld by the native burn/recycle rule [45]. Pin its UID, registration generation, owner association, and the burn setting before weight submission; validate them again for the payout cycle. The live configuration must admit a single nonzero weight and a full-weight destination. The latter weight cap is root-controlled; owner control alone cannot establish this precondition [46]. Failure to meet these conditions blocks launch of this adapter. Do not substitute zero weights or leave a previous earned vector active.

Withheld owner-directed incentive can reduce the subnet's later emission share. Burning is therefore an economic cost, and a zero-incentive epoch has separate native redistribution behavior [47]. Quantization, registration changes, native reveal lag and the fallback must be demonstrated against the pinned runtime. A frozen application certificate specifies the intended weights; native consensus still determines the resulting payouts.

Execution and incentive limits. The two native task pools above reward independently evaluated A artifact performance and B output utility. Public A models can still favor colluding B outputs, and public weights make copying possible. Qualification, duplicate handling, execution budgets and actual native settlement must be tested before live rewards. The incentive study must include miner incentives and validator dividends received by a common owner, including native filtering, normalization, bond state and weight-reveal delays. Application-level credits alone do not establish that owner's native payout. The design does not claim to eliminate all profitable self-dealing.

9 Style measurement and final review

This section defines the style utility V and semantic gate G for B transformation. Every B task requires preservation of the source's full meaning, fulfillment of the task's target tonal intent, and readability. These constraints create no separate writing-improvement service or reward pool. Positive utility requires independent semantic certification; style proximity or detector evasion cannot compensate for a preservation failure. Numerical measurement uses a public frozen artifact. Semantic judgments are explicit oracle inputs certified under the validator-weight assumption in Section 10.

Validators obtain authoritative benchmark origin probabilities by executing the published A panel under the artifact-freeze and execution rules in Sections 4.1 and 8.2.3. They compare B's submitted vectors with canonical outputs bound to the same artifact, runtime and input. B models remain private; rewards assess their committed output text. Every credited B entry also requires its version-bound execution receipt under Section 4.3; that check cannot substitute for any semantic field. B miners can compute public A scores during search, while hidden labels and unreleased semantic-review evidence remain restricted.

Frozen style measurement. Style measurement is required for every B benchmark, alongside Pangram and the frozen subnet detector panel. Each task supplies authorized target-author references and a frozen style calibration before issue. Before task issue, publish a permitted word/grammar author-distance artifact and pin its hash, parser and feature schemas, model parameters, reference-pooling rule, executable/runtime, numerical operations, rounding, resource bounds, and test vectors. Its distance function $d_\theta(t, R)$ returns a nonnegative integer for text t and target-author references R. The instance manifest must supply these concrete bindings; this specification does not select or claim to have trained the artifact. Private B generator weights do not define this public style evaluator.

Freeze a nonempty calibration multiset $C = (c_1, \ldots, c_M)$ of integer distances from development queries to their attributed authors' independent reference works. Keep development source families separate from fitting and evaluation, record provenance and hashes, and use the same artifact and reference-pooling rule. The task pins its calibration ID before candidate generation. Retain duplicate distances. With $Q = 1{,}000{,}000$, define the integer style fit

$$q(t, R) = \left\lfloor Q \, \frac{2\#\{m : c_m > d_\theta(t, R)\} + \#\{m : c_m = d_\theta(t, R)\}}{2M} \right\rfloor . \tag{9}$$

This empirical rank gives exact ties half credit. Smaller distance cannot reduce q. It is a calibrated distance rank, not an authorship probability or a certificate of preserved meaning.

For source x and candidate z, evaluate both against the same R, artifact, and calibration. Let $q_x = q(x, R)$, $q_z = q(z, R)$, and define

$$V(x, z) = \begin{cases} 0, & q_x = Q, \\ \dfrac{\max(0, q_z - q_x)}{Q - q_x}, & q_x < Q. \end{cases} \tag{10}$$

Equal fit earns zero improvement; a saturated source has no remaining headroom. Preserve the exact rational numerator and denominator through aggregation. Every B comparison requires $q_z \geq q_x$, even though V clips negative improvement. Task mixtures and final credit rounding remain part of the frozen allocation policy.

Score the source and references successfully before issuing a B task. Any candidate that receives the declared deterministic `unscorable_candidate` status earns zero B utility, since style nonregression cannot be verified. Execution disagreement or unavailable shared evidence cannot silently supply a score: retain an unresolved status and withhold credit under the candidate-specific or certified shared-scope rule below. A numerical failure alone does not establish attributable service nonresponse. Validate the frozen artifact against copying, topic substitution, and meaning-breaking controls before funds depend on it.

Miners can also optimize directly against the public style artifact. Before live rewards, test adaptive searches that maximize V under Section 14. Among revisions that pass P, R, T, higher V must still track independent reader judgments of author fit; the semantic gates alone do not validate that ranking.

Three semantic fields. The source's full meaning must remain intact. Before generation, resolve the target tonal intent from the brief and, where supplied, the requested authorial style and authorized reference samples; freeze it in the task rubric. A dramatic change of authorial style can authorize a different tone without a separate tone instruction. Use source tone as the default for aspects the request leaves unchanged. Review the whole candidate for meaning, including information not enumerated in the checklist. Tone review applies the frozen rubric and its source-tone defaults throughout the candidate; reviewers cannot add new stylistic preferences after generation. Every certifying validator records one verdict, `PASS`, `FAIL`, or `ABSTAIN`, for each field:

- P, preservation: retain the source's full meaning, including all claims, entities, quantities, negation, uncertainty, attribution, citations, and argument relations. Changes to expression must preserve that information. Unsupported material additions, omissions, altered meaning, or prohibited reference copying fail. Greater stylistic force cannot strengthen factual certainty or obligations.

- R, readability: introduce no material readability regression against the source and satisfy the brief's stated readability requirements, including required cleanup of identified unwanted filler, repetition or formulaic phrasing. Shorter text or simpler vocabulary alone supplies no pass.
- T, tone: fulfill the frozen target tonal intent and every tone and audience constraint. Preserve the intended voice features, including informality or idiosyncrasies, specified by that target. This field always requires a semantic vote, including when no separate tone instruction appears in the request.

Insufficient evidence or unresolved interpretation, including uncertainty about the target tonal intent, requires abstention. Failure ballots identify rubric items and bound source/candidate spans where applicable. Preference ties and extra editorial praise earn no additional utility. No detector probability substitutes for these semantic fields.

Authority and timing. Three hash-assigned preparers assemble the dossier; their vote count cannot finalize a judgment. Every final signer receives authorized benchmark evidence, checks the relevant source, references, brief, candidate and rubric, and attests the semantic field itself. The review client presents semantic evidence first; honest signers commit their field ballots before inspecting detector scores or numerical style results. Full-set authorization is not a cryptographic blinding guarantee. Checking three preparers' signatures or tally is insufficient. Reuse the verified frozen validator snapshot and integer effective weights from Section 10, including its conditional assumption of strictly less than $W/3$ dishonest eligible weight. Positive W, qualified recipient keys and complete capacity reservations are required before issue.

For the pilot epoch starting at H, preparers commit by $H+180$ and open privately by $H+192$. Evidence corrections close at $H + 196$; they may correct references to already committed evidence, but cannot change the candidate, brief, evaluator, or calibration. Each validator then commits its one final ballot per field by $H + 198$, opens it privately by $H + 202$, and the global field certificate must be included by $H + 204$. Deadline equality is timely; finalized inclusion, incident scope and payout timing follow Section 10. Reserve all witness publication and verification work before issue. Under the pilot response schedule, candidate certification occurs by $H+72$ through $H+168$, leaving 30 to 126 blocks before the ballot commitment cutoff: approximately 6 to 25 minutes at 12 seconds per block. Certifiers must review the authorized source and candidate as evidence becomes available; waiting for the preparers' private opening would leave only six blocks. These cutoffs do not establish measured review throughput.

The first canonical timely commitment fixes that validator's ballot for the field. Identical retries are idempotent; a conflicting signed ballot is retained as evidence and cannot replace it. Late, absent, or invalid openings contribute no agreeing weight. Keep missing, abstaining, recused, and nonresponsive validators in the frozen denominator W. A `PASS` or `FAIL` certificate requires distinct eligible signers of that verdict with total weight w satisfying $3w > 2W$. Honest signers never authorize conflicting final verdicts; conflicting strict certificates halt affected settlement.

Deterministic closure. At the close, set $G = 1$ only if all three fields have a valid `PASS` certificate. Set $G = 0$ if any field has a valid `FAIL` certificate. Otherwise the judgment is unresolved. Conflicting certificates take precedence and halt settlement. Missing candidate submissions remain attributable miner nonresponse under the service-evidence rules.

At the fixed close, an unresolved candidate contributes zero earned credit for its assigned slot. Retain its unresolved status and unknown G; do not invent a `FAIL` certificate. Keep that slot's preassigned weight in aggregation. Other candidates with complete evidence retain their independently earned credit. Candidate-specific numerical, receipt or semantic uncertainty cannot erase their results. Do not draw replacement judges or allow a replacement candidate to obtain a preferred verdict. An unresolved semantic judgment adds no service failure or recovery deficit; independently attributable service failures remain.

A common source, frozen evaluator or certified platform failure can make shared evidence unavailable for every entry that requires it. Apply the same zero-credit/void treatment to that entire declared scope and retain its assigned aggregation weights. Establish the shared cause and scope under the frozen certificate rules; a candidate-triggered error or a minority allegation cannot become a global exception. In the absence of such evidence, handle unresolved results individually. Conflicting strict certificates still halt affected settlement.

Selective non-certification can withhold a valid candidate's credit. A dishonest minority below $W/3$ cannot prevent a certificate when all remaining weight agrees and participates, but semantic disagreement or honest outages can prevent that agreement. Publish candidate-level coverage, abstentions and reviewer patterns. Validate review reliability, candidate-triggered failures and owner-level payoff under the bounded study in Section 14. Quorum agreement does not prove semantic correctness.

The versioned adjudication record binds the task and comparison IDs, source/candidate/reference/brief commitments, evaluator and calibration hashes, exact q_x, q_z, V, rubric fields, evidence root, validator snapshot and W, ballot commitments/openings, field certificates, deadlines, G, and any unresolved or void reason. Records use the canonical encoding and authenticated encrypted evidence paths of Section 10. Evidence must reach every authorized certifier within the funded publication limits. Customer jobs never enter this process automatically. A execution records additionally bind the complete published model manifest and canonical runtime configuration. Every credited B record also retains its verified receipt, instance evidence and accepted policy. Certificates attest scoped judgments and faithful computation. They do not prove semantic correctness or require reproduction of private B generation. A artifact loading, numerical replay, and full-matrix capacity require their own implementation checks beyond the one-ticket A-requester/B-provider rewrite mailbox pilot.

10 Mandatory-service evidence and closure

The proposed `service-evidence-v1` mailbox covers bounded, protocol-assigned rewrite requests from A requesters to B providers. A models are public; B generators remain private. The requester counters, provider assignments and availability gates are defined in Section 8.2. The one-ticket exercise below specifies transport and review of a B rewrite. It implements neither an artifact runner nor the complete common source/candidate matrix. An emission-bearing benchmark needs a separate frozen execution schedule and capacity budget. Emission-bearing B service also needs a qualified-version execution receipt under Section 4.3; the transport pilot alone cannot establish that requirement. Private customer jobs use a separate instance-key and receipt flow and never enter mandatory-service counts. Their plaintext is restricted to the customer and verified runtime. Curator-authorized witnesses below may disclose benchmark text to the listed providers and validators, but must never export customer keys or override the customer boundary. Full encrypted payloads are published on chain; a hash or HTTP locator alone is insufficient. Publication establishes bytes, sender authentication, and timing. Private validity and quality require the certificates below. This mailbox is a specified pilot, not an implemented throughput result.

Authoritative A execution follows Section 4.1; the common benchmark matrix and its separately funded capacity follow Section 8.2.3. The mailbox certificates below establish transport and declared service validity. They do not replace benchmark execution or semantic review.

Frozen authority. Before revealing task contents, exclude the exact assigned A/B UIDs from the round's eligible validator set. Bind every remaining UID, registration generation, hotkey, signing and encryption key versions, and integer native effective consensus stake, including native alpha/TAO scaling, to one finalized state snapshot. Its manifest names the block hash, runtime and storage schema, weight field and units, complete records, and total weight W. Initial loading and later snapshots require verified finalized state proofs or a chain-verified native snapshot. A relayer's assertion alone is insufficient [48].

Every final validity, quality, or incident verdict requires signatures whose distinct frozen weights sum to w, with $3w > 2W$. Assume dishonest weight is strictly below $W/3$ in this conditional set. Missing keys, nonresponse, recusal, and later key changes never shrink W. Three hash-assigned preparers organize evidence; they cannot authorize a verdict. Honest validators independently check evidence and sign at most one verdict per stage. Conflicting strict quorums would share more than $W/3$ weight; a detected conflict halts affected settlement. Signatures bind the verdict itself, evidence, snapshot, stage, and scope.

Fixed pilot. The instance manifest supplies actual chain/contract addresses, verified snapshots, authorizations, service/schema hashes, and funded reservations. Missing bindings, zero eligible weight, or insufficient qualified key and verification capacity prevent issue. At native epoch start H, issue at most one authorized rewrite ticket from an A requester to a B provider in a 360-block measurement epoch, with three prescribed B providers total. Freeze these settings before issue:

Parameter	Value
Input / output cap	4,096 / 8,192 bytes
Ciphertext chunk / publication cap	4 KiB / 8 MiB per stage
Global epoch publication cap	64 MiB, including all witness copies
Request publication / certificate	$H + 12$ / $H + 24$
B provider response window	36 blocks
Certificate window per attempt	12 blocks
Preparer commitment / private opening	$H + 180$ / $H + 192$
Final verdict and incident close	$H + 204$

Attempt $j \in \{1, 2, 3\}$ starts at $H + 24 + 48(j - 1)$, publishes by start plus 36 blocks, and certifies by start plus 48 blocks. The final B attempt cutoff is $H + 168$. Early declines do not advance later windows. Reserve all possible bytes, gas, response and verification capacity; insufficiency prevents issue and does not authorize another draw. One ticket provides a bounded transport pilot, not availability evidence for every UID. Settlement names a later native payout cycle and its actual commit/reveal configuration; no claim is made that reveal delays fit within the unused part of this epoch.

DATA, then witness. Use HPKE base mode with X25519/HKDF-SHA256/ChaCha20Poly1305, fresh single-shot ephemeral material per recipient, and separate registered sender signatures [36]. Deterministic CBOR records reject duplicate keys, indefinite encodings, tags, and floats [49]. A salted payload commitment and encryption context bind protocol, chain, contract, ticket, stage, attempt, both UID generations, and recipient key version. Input DATA copies address the requester and three prescribed providers; response copies address the requester and actual provider. Curator authorization covers those recipients and the complete validator set. Hidden labels and review answers remain in validator-only evidence packets.

First fix and publish the signed DATA manifest and all ciphertext chunks. Then form a witness containing that DATA digest, plaintext, salt, and each DATA envelope's ephemeral generation material. Encrypt the witness separately to every frozen validator and publish those complete ciphertexts under a second manifest referencing DATA. Validators reconstruct each DATA encryption and compare the exact bytes and recipient bindings, as well as authorization, commitment, and schema. Decrypting only their own copy would not check the miner's copy. DATA never references a witness hash; witnesses do not contain their own encryption secrets. This avoids a hash cycle. Generation witnesses remain confidential and need implementation/security review. Their certificates do not assert that every validator decrypted its witness copy. Public ciphertext remains exposed to future key compromise; authorized archives retain complete replay evidence.

Transitions and attribution. Issue permanently consumes the requester index and activates its obligation. One immutable manifest fixes each stage; identical retries are idempotent and conflicting signed manifests remain evidence. The requester's decline, absent complete publication, or certified invalid input is its failure. An authenticated decline included by the stage's publication cutoff is terminal for that stage. Retain later bytes as evidence, but reject an acceptance certificate that ignores the decline; late declines cannot reopen closed accounting. Only a timely global input-acceptance certificate activates the first provider at its fixed start. No input acceptance means no provider obligation. The provider's decline, absent publication, or globally certified invalid response permits the next fixed fallback; successful fallback never erases preceding failures. Unresolved validity stops dependent work without another draw. Late answers remain late.

Complete publication without a global validity verdict closes unresolved; recipient accusation alone cannot assign fault. Keys must be qualified before the snapshot. Key loss afterward cannot blame a sender whose encryption was certified correct; rotations and UID reuse cannot change issued tickets. Certified delivery of a B response does not establish its revision quality. A request-publication failure remains in its requester ledger and cannot alter a benchmark artifact or computed result. Validator runner failures never enter this rewrite-service ledger. Missing service-validity certificates do enter it as unresolved obligations, which block epoch eligibility even without miner fault (Section 8.2). B rewards require a qualified generation receipt and independent review of the submitted revision, without validator replay of private weights. Insufficient candidate-specific review evidence withholds only that entry's credit; certified shared failures affect their complete scope. Retain assigned weights and independently attributable service failures under Section 9. The companion service-evidence specification [50] defines the detailed record and certificate rules.

Counts and finality. For each applicable A-requester or B-provider class, retain activated obligations $A = S + F + U + V$: success, attributable failure, unresolved, and platform void. Unused reservations are separate. Let $N = S + F$; require $U = 0$, $N > 0$, and $10F \leq N$ for the class and aggregate availability gates. Publish all counts. Keep $D_e = \max(0, D_{e-1} + 10F - N)$; neither unresolved nor void work retires a deficit. Require zero outstanding deficits and current qualification; customer jobs and other classes cannot dilute these gates. A artifact publication and qualification are separate eligibility gates.

Only canonical finalized inclusion counts, with deadline equality timely. Subtensor separates block production from GRANDPA finality [51]. A local timeout or censorship allegation does not prove a platform exception. Normal operation assumes fair inclusion and sufficiently prompt finality. A finality stall delays observing closure without extending block-height deadlines. A strict global incident certificate must be included by $H + 204$, even if its block finalizes later, and identify the affected interval and every intersecting ticket/comparison. The rule voids all activated obligations in that scope, including successes and failures, and cannot selectively pardon a miner, shrink W, or reopen closed epochs. Without that certificate, normal inclusion rules apply; missing verdicts remain unresolved. Gas, history retrieval, proof-verified snapshots, and the declared native settlement timing must be demonstrated before this EVM pilot issues work [38].

11 Detector evidence and assigned audits

Pangram's API provides asynchronous tasks, analyzed text, document fractions, version information, and optional public dashboard links. The returned text may differ through preprocessing, so validators must bind evidence to the analyzed bytes rather than a preview. The local reader currently requires exact equality; a normalization policy would need its own version and review. Pangram API [52].

Our synthetic probe obtained a publicly retrievable report containing the analyzed text and scores. An independent caller retrieved it without a new paid inference. The observed web endpoint is undocumented and needs an availability, retention, and usage agreement before production dependence. We have not established a provider signature, immutable model identity, or one unique canonical result per text. Recorded probe [53].

For every B benchmark, a miner commits its final candidate and required report IDs before the evidence deadline. Every validator in the round's frozen certifying set receives the source, candidate, report URLs, score projections, and commitment openings in a separate encrypted evidence envelope. Each signer must retrieve the provider's record and inspect the exact text, fractions, version, time window, and required preservation evidence; preparer signatures cannot substitute for that inspection. Report IDs and URLs remain encrypted in chain payloads and absent from public plaintext logs. Anyone who obtains a public URL may still retrieve its content; encrypting the locator protects our delivery path, not Pangram's hosting. Curator authorization must cover all frozen certifiers. Private customer jobs remain outside this benchmark; a customer's optional disclosure to one chosen validator does not authorize wider benchmark disclosure.

Pangram states that it does not train Pangram 4 on customer API data. Private B weights prevent direct downloading, but do not prevent independent black-box study through purchased outputs or leaked examples. Attested inference does not prevent authorized recipients from retaining or studying outputs. Provider verification and any reuse of provider labels also need suitable service terms. Pangram model card [12].

Public A artifacts can be inspected by Pangram and other providers. Pangram remains an independent external origin check when a subnet detector is unavailable, but its result cannot stand in for author attribution, preservation or a missing A result in the joint benchmark. Report external and subnet-detector outcomes separately; missing A evidence cannot become an adversarial pass. Customer text is not sent to Pangram without the customer's separate authorization.

For public A scores, use the complete comparison matrix, B self-score bindings and signed execution records in Section 8.2.3. Every final signer independently verifies the required computations under Section 4.1; a coordinator's signature alone is insufficient. For every credited B entry, also verify the private-version execution receipt under Section 4.3; it does not substitute for a Pangram report. Benchmark authorization permits these external reports. Customer attestation never authorizes an external call. Section 10 specifies complete on-chain ciphertext publication, separate confidential generation witnesses, and fixed validity/closure certificates. A report URL, response hash, or recipient accusation alone cannot satisfy that service-evidence contract. Its runtime implementation still needs validation independently of the report reader.

11.1 Quality anchor, Pangram cadence and parity

At launch, schedule one joint B benchmark round in each scoring epoch. Freeze its admitted miner cohort, comparison coverage, report budget and deadlines before issue. Every credited B candidate requires both the author-fit evaluation and the complete Pangram/subnet evidence defined in Section 8.4. The submitting miner funds its three candidate reports; the benchmark operator funds the three-report shared source baseline and separate parity studies. Reserve full validator inspection and execution capacity before issue. The 24-task pilot must test feasibility; it has not established that this workload fits the declared epoch.

This cadence governs scored benchmark submissions. Miners can train and search locally between rounds using public A artifacts. Private customer text retains its exclusive-recipient policy. A prior report, another candidate's score or an untested estimate cannot fill a missing Pangram observation. Apply the existing evidence and closure rules when required reports are unavailable, and apply current service-availability and recovery gates at settlement.

In each A scoring epoch, compare all admitted A origin artifacts and the incumbent quality anchor with Pangram on the same fresh, independently labeled hidden origin set. The operator funds the shared Pangram observations and reserves their budget before issue; this cost does not multiply with the number of A artifacts. Freeze the reference set, provider version, repeat policy, operating points and analysis before evaluation. Publish performance gaps between A and Pangram, uncertainty and coverage, including missing provider evidence. Missing evidence cannot support a current parity finding. Reuse a B report only when its exact text and observation satisfy this independent evaluation protocol; a set of miner-selected successful evasions cannot replace the labeled reference set. Continue measuring Pangram when a subnet detector holds the anchor role.

Pangram is the initial anchor. A qualified subnet detector takes over after demonstrating superiority to the incumbent; a later Pangram improvement can make Pangram the target again. Before opening evaluation data, freeze the incumbent, contender versions and a primary comparison statistic based on detection sensitivity at the declared human false-positive operating points. Freeze population weights, required slice limits, a practical superiority margin and uncertainty analysis. Use model-only or mixed production as the positive event and human-only production as the negative event for both detectors. A contender's improvement over the incumbent must exceed the margin without violating the limits. Nominate the highest-scoring admissible contender before opening fresh confirmation data. Confirm its superiority on that unreleased set; account for selection among multiple artifacts in the analysis. Ties, inconclusive results and missing evidence retain the incumbent. This rule selects the strongest validated reference under the declared comparison policy; the B panel's origin-Brier ranking can select a different detector.

Certify each anchor change under the frozen validator quorum, binding its artifact or provider version, comparison policy, evidence and future activation epoch. A switch cannot alter an issued task, its reward normalization or its assigned detector panel. Provider updates and observed performance changes prompt renewed comparison; a provider announcement alone cannot change the target. Anchor selection leaves independent production histories as ground truth and preserves the fixed public probabilistic baseline used for A reward normalization.

Reducing B's per-candidate Pangram checks requires demonstrated detection parity. Before opening the evaluation set, freeze the actual A champion and panel policy proposed for substitution. Compare them with Pangram on the same independently documented human-only, model-only and mixed production histories, including fresh adversarial B revisions and held-out writers, source families and registers. Keep those production labels after successful evasion. Measure detection sensitivity at declared human false-positive operating points and require the frozen human false-positive limits to hold, including predeclared difficult slices. Fix noninferiority margins, sample sizes, report aggregation and uncertainty calculations before collecting results; confirm the finding on successive unreleased evaluation sets. Agreement with Pangram alone is insufficient.

Pangram's content fraction and A's document-origin probability have different meanings. Any calibration comparison must first map them to the same labeled event using separate development data. Bind parity findings to the tested versions, policy, populations and dates; changes in provider version or observed deterioration require renewed evaluation. Publish aggregate results and certify the finding under the frozen validator quorum.

This launch version retains Pangram in every scored B round, including after a parity finding. Parity permits consideration of a future policy with less frequent external checks on B candidates. That policy must retain periodic B checks and A's continuing Pangram quality comparison, and specify its interval, scoring of unchecked candidates and response to drift before activation. It may not alter an issued comparison or claim a Pangram pass for text the provider did not check. There is no automatic switch based on miner claims of parity.

11.2 Assignment and audit sequence

A training requesters and B providers are paired by the lifetime interaction counter and frozen eligible roster. Benchmark batches, common detector panels, and preparer assignments use a separate, shared comparison counter and distinct hash domains. These schedules are publicly predictable. At launch, review every scored submission and all required checks, including whole-revision preservation. There is no hidden audit sample and no application-level external beacon dependency.

For each scored submission, rank eligible validator UIDs by the hash of the versioned review domain, chain, subnet, shared comparison index, submitting UID, and candidate preparer UID. Use canonical encoding, UID tie-breaking, three distinct evidence preparers, and an ordered reserve list. Exclude the exact miner UIDs involved in that comparison before freezing its certifying validator set and native effective-consensus-weight snapshot. The coordinator cannot choose preparers or obtain another draw; different UIDs can still have the same owner. Reserve preparation and full-set verification capacity before issue.

Preparers organize checks and retain independent commitments/openings. Final certificates follow the frozen authority and strict weighted quorum in Section 10. They attest the actual verdict; preparers have no separate approval authority.

1. Freeze the eligible epoch roster from qualification and actual request outcomes, service capacities, global task budgets, index-allocation rules, deadlines, and authorized certifying validator roster. Verify its native weight and identity records against the finalized snapshot specified in Section 10. Commit the question pool and source Pangram baselines. Publish fixed block deadlines for submissions, complete evidence, preparer commitments, private openings, dispute resolution, and weight submission.
2. Issue A-requester/B-provider rewrite tickets in the protocol schedule. Consume consecutive lifetime indices, derive their ordered peers from the pinned roster, and record responses, declines, expiries, and fixed fallbacks. Training feedback uses released, authorized material; active benchmark labels and detailed review material remain private.
3. Use the shared comparison index to determine the common A panel and B batch under the frozen policy. Exclude every B UID from A reserves before freezing pre-issue artifact reserves. Validate and cache the complete selected A artifacts, reference runtimes and thresholds before revealing the B task or starting generation. Accept one final B candidate and selected Pangram-report commitment per assigned task. The predictable opponent schedule does not allow a second draw or a revised submission after commitment.
4. Execute the required common panel matrix in the restricted reference runner, retaining control outcomes and execution failures. Apply Section 4.1 for runner recovery and Section 9 for unresolved computations. Record any missing strongest-detector comparison. Finalize all evidence commitments before the fixed evidence deadline; a favorable score is never grounds to switch provider or discard an observation.
5. Publish complete recipient-bound DATA ciphertexts, followed by separate confidential witness/evidence envelopes to every frozen certifier, under Section 10. Retain finalized inclusion proofs and enforce key versions, expiry, replay state, and publication deadlines. Preparers commit answers before peer openings, then open privately to the frozen certifiers. Every signing validator inspects the evidence, verifies the DATA-generation witnesses and report/panel bindings, and signs the exact verdict under the strict weighted threshold. Close unresolved records under the fixed rule before the declared settlement cycle; missing evidence never grants approval. Withhold hidden labels, other miners' candidates and detailed review material until retirement. B can calculate its own public A scores throughout local training and search.

Retain every review and disagreement. A missing preparer follows its fixed reserve policy; a missing global certificate closes unresolved under Section 10. Neither defaults to approval. Full coverage prevents a miner from concentrating defects in a publicly predictable unreviewed subset. It does not establish validator honesty or guarantee that readers detect every error. Weighted certificates require the stated honesty bound, and stake identities alone do not establish it. Reduce task volume to fit the funded preparation, ciphertext publication, and full-validator inspection budget instead of reducing coverage or the certificate denominator. Any later sampling policy needs a separate security argument and measured cost benefit.

Subtensor's timelocked commit-reveal protects validator weights from immediate copying. Use its supported SDK/runtime path, including any beacon dependency within that native mechanism; this proposal adds no separate beacon wait for peer allocation or auditing. Weight plaintext eventually becomes public; customer inputs must not be encrypted with that eventual-reveal mechanism. Use the recipient encryption described above and do not hard-code a CRv4 schedule. Bittensor commit-reveal [54].

12 Trust assumptions and adversarial behavior

The chain can enforce authenticated accounting and commitment consistency. Recipient encryption limits who can decrypt protocol deliveries. Attested jobs additionally restrict key release and plaintext processing to the accepted runtime under explicit hardware assumptions. Counter and roster commitments permit reproducible assignment of peers, comparison batches, and reviewers. Every scored submission receives the required review at launch. Each guarantee has a narrower scope than useful, confidential writing inference.

Risk	Proposed response and remaining assumption
Customer text reaches another miner or gateway	Encrypt locally to the pinned assignment; no alternate keys or automatic reassignment
Ordinary A host leaks decrypted text	This mode grants operator access and requires customer acceptance; paid B requires attestation
Host reads B text, logs or GPU buffers	Attest the complete CPU/GPU path and approved application before customer key release; disable debug, logging, external tools and operator administration
Customer or separate compute host obtains private B weights	Release model keys only to an approved loader; expose no weight or debug interface. Paid-query extraction and behavioral imitation remain possible
Malicious miner code runs in a real enclave	Public reviewed runtime and measured policy, constrained private model data and denied egress. Hardware identity alone does not approve an application
Stale quotes or substituted keys redirect requests	Bind fresh customer nonce, instance keys, offer, model and policy; check revocation and refuse downgrade or alternate recipient envelopes
Private model or shared cache leaks another customer's data	Isolate job state, prohibit online training, cache sharing and exported state; malicious-data and timing attacks still require evaluation
Hardware or verifier is compromised	Pin accepted roots and versions, independent security review and revocation before new key releases; previous exposure cannot be undone and side channels remain a limit
B miner retains private weights but fabricates quality claims	Score submitted revisions against the brief and independent review; every credited B entry requires a verified receipt for its qualified version and bounded serving profile
B supplies false self-scores	Validators execute every assigned A artifact on the bound source and final text and compare canonical vectors and scalar projections; B's report does not authorize its own reward
Miner copies reference content or removes difficult claims	Full preservation review, copying checks, and explicit exploit controls

Risk	Proposed response and remaining assumption
B miner chooses a candidate that prevents a semantic quorum	Withhold that candidate's credit while retaining its assigned denominator and competitors' completed scores; selective non-certification remains a risk
Miner's private search and retry costs are unverifiable	Miners fund their own attempts; score committed reports under fixed deadlines; do not reimburse claimed private costs
Validators collude or score inconsistently	Every final verdict needs strictly more than two-thirds of frozen eligible weight. Assume less than one-third is dishonest; every signer inspects the evidence and missing votes remain in the denominator
Curator or coordinator leaks labels or selects a biased task set	Independent provenance review, overlap checks, and source-group splits; commitments constrain later changes to the frozen pool and baseline evidence. Label secrecy and honest curation remain trust assumptions
Published A detector favors a transformation miner	Hidden qualification, targeted-trigger tests and a common frozen panel executed by validators; replay verifies computation but not the absence of colluding model behavior. Distinct UIDs do not prove independent models or owners
Model content is copied under additional identities	Pin complete artifacts and give identical inference content one model-credit entry and panel seat; small changes or a new hash do not establish an independent contribution or owner
Pangram changes or serves conflicting records	Pin the reported interface/version, retain observations, and suspend affected comparisons under a fixed policy
Customer consumes B work and withholds acknowledgment	TEE completion receipt, complete ciphertext publication and strict delivery certificate release payment without customer acknowledgment
Miner claims B delivery with a receipt or URL alone	Require the exact complete customer ciphertext on chain and a bound valid runtime receipt; validators cannot waive contract checks
Paid validators withhold certification	Fixed roster and deadlines; certification timeout refunds the customer even after valid delivery, leaving bounded miner risk under quorum and chain-liveness assumptions
Miner manufactures paid demand	Keep customer revenue separate from emission scoring

Attestation adds trust in hardware vendors, firmware, the accepted workload and the customer's verifier. It supplies no proof of semantic fidelity or independent UID ownership. Customers retain control of their own accepted policy; protocol signers cannot release customer keys on their behalf.

The quorum assumption applies after excluding the exact task participants; it must hold in that conditional validator set. It prevents conflicting certificates when honest validators sign once. It does not prove their semantic judgments correct or a published detector accurate. Authoritative A outputs come from validator execution of the frozen artifact, with evidence checked under the declared runtime and output rules. The quorum still assumes enough honest validators perform those checks. Native Yuma also processes each validator's submitted weights independently. A valid benchmark certificate cannot force a dishonest validator to submit its prescribed weight row. Public replay establishes the executed computation. Native payout analysis must also account for validator weight choices, bond state and any miner and validator receipts controlled by a common owner.

The benchmark also needs private source rotation, duplicate detection, and delayed release of retired examples. Published detectors may recognize a repeated test or contain a trigger shared with B. Under Section 8.2, B can inspect public A weights and compute scores for local optimization at its own cost. Those probabilities are not concealed feedback and do not validate the resulting text or its provenance. Changing the artifact, benchmark or judge after observing a candidate would invalidate the declared comparison. Publish protocol versions and aggregate outcomes while keeping customer material, hidden labels, unreleased sources, other miners' candidates and private review evidence restricted.

Training tickets bound aggregate B rewrite work requested by A. The predictable hash schedule does not prevent common owners from obtaining multiple UIDs or exploiting occasional favorable assignments. It provides verifiable allocation without an ownership registry. Service failure rates need authenticated request and delivery evidence; the strict greater-than-10% rule cannot be based on a recipient's complaint alone. Predictable assignments remain mandatory, and requester abandonment consumes both an index and its failure allowance. The permitted 10% allows selective refusal. Timely valid rewrites may still fail quality evaluation; availability does not certify quality. The 90% gate and recovery deficits apply to issued A-requester and B-provider obligations. Separate role counts prevent cheap successes from diluting failures. A persistent recovery deficit makes abandoned work delay later eligibility even after the current epoch's credit is lost. It cannot make an operator who plans to leave value future participation. Cache and validate the frozen A artifacts before task issue. A runner outage can use reserved capacity for the same artifact; an unresolved reference execution requires the declared common-comparison closure and cannot select a different detector after seeing B's candidate. A has no required inference endpoint. Training-service qualification uses actual A-requester and B-provider obligations and deterministic probation work, without heartbeats. Optional paid hosting has its own customer terms.

13 Existing evidence and implementation status

The current Rust representation uses word occurrences and grammatical features from spaCy annotations. It scores complete sparse distributions and learns selected coordinate corrections. It does not require an LLM to enumerate or measure deterministic edit candidates. The current author/detector models and editing rules support experiments for A detection and B transformation; a demonstrated end-to-end product remains to be built.

The table reports top-1 accuracy averaged within each author, then equally across authors, then across three seeds. Query counts describe unique inputs scored in every seed; these author-macro percentages are not pooled fractions of those counts. Evaluation protocols [55,56].

Experiment	Author-macro top-1	Scope
Blog author retrieval	56.30%	600 evaluation authors in six 100-author galleries; 1,206 queries
Global Voices transfer	73.81% combined; 77.98% lexical-only	Fourteen provisionally qualified authors; 58 queries; topic/editorial confounds
Compact document projection	32.82% Blog; 25.00% Global Voices	Development-selected linear encoder; previously evaluated test corpora
Selected-feature ablation	40.83% Blog; 29.17% Global Voices	Conditional diagnostic retaining the frozen scorer's pooling and calibration

The Global Voices comparison removes grammar terms from the frozen Blog-trained scorer without refitting or renormalizing any remaining term. That removal improves accuracy by 4.17 percentage points on this sample, reversing the grammar contribution's benefit on Blog data. The experiment does not identify why: it has no matched-topic control and does not isolate register, editorial practice or extraction effects. This supports testing feature weights across registers; it does not establish that grammar is generally unhelpful [56].

The projection experiment shows why 3,557 learned coordinate weights must not be mistaken for a sufficient fixed input vocabulary. The full model also retains unselected word and grammar contributions. Neither author retrieval nor these feature ablations establish meaning-preserving editing or detector evasion. In a separate edit diagnostic, replacing *may* with *must* improved author proximity despite changing the claim. Independent preservation checks are therefore central to the design. Author projection study [55], edit counterexamples [57].

Implemented components include Rust tools for corpus collection and evaluation, portable author scoring, offline challenge and submission checks, a public Pangram report reader, and an HPKE prototype that encrypts receipts for assigned validators. The receipt prototype relies on caller-supplied trusted assignments; it is not a customer-request protocol or deployed key registry. Receipt implementation [58].

The project does not yet demonstrate a reliable sub-10% rewriting model, calibrated production-origin service, reviewed commercial training release, authenticated live benchmark ledger, customer escrow contract, or private serving endpoint. The results justify a measured pilot and constrain its design; they do not establish readiness for emissions or customer funds.

14 Development and launch criteria

First, run a local tournament with 24 rights-cleared tasks across the two launch interfaces, A detection and B transformation. Evaluate detection on known-label author and origin tasks against fixed probabilistic baselines. For transformation tasks, compare unchanged text, deterministic edits, and one general editor under the same briefs, target-author references and Pangram/subnet detector panel. Measure both objectives on every B benchmark; isolated objective ablations are diagnostics and earn no B emission credit. Include scored human-only, model-only and mixed sources within this pilot's B allocation, with origin mixture and weights frozen before issue. Include a human-only source whose required Pangram reports all score zero to verify that style improvement can earn credit only while every required candidate report remains zero and subnet nonregression and semantic gates pass. Report outcomes by source origin as well as in aggregate. Three hash-assigned preparers assemble dossiers, including intentional omissions, uncertainty changes, and padding. Frozen validators independently assess B's preservation, readability and tone; strict weighted certificates finalize those fields. Preference remains a diagnostic. Freeze the public style artifact and calibration before comparison and publish aggregate agreement, coverage, cost, and failure cases. Reserve and measure the per-epoch joint B benchmark and continuing A quality-reference schedule, including operator-funded Pangram observations, in Section 11.1. Test that no B credit can omit author references, style calibration or required detector evidence, and that raw regression on either objective cannot be hidden by a clipped score. A parity study must use independent production records and the frozen comparison protocol before any proposal to reduce B's Pangram call frequency. A's external comparison continues after parity and after any such B policy change. Exercise anchor changes from Pangram to a superior A artifact and back to an improved Pangram. Require independent confirmation under the frozen comparison policy; ties, uncertainty and missing evidence must retain the incumbent. Verify that a switch takes effect only in a future epoch and cannot change an issued task or its reward baseline.

Evaluate cleanup and author consistency through blinded comparisons with held-out writing in the same register and, where available, human revisions under the same brief. Record removal of unwanted patterns, retention of intentional voice features, and preservation failures separately from detector scores. Before claiming indistinguishability, predeclare the comparison task, acceptable difference margin and sample size, and report uncertainty intervals against that margin. Failure to find a significant difference in a small sample is insufficient.

The 24-task pilot intentionally prioritizes protocol and failure measurement. Reserving four B comparisons for the attacks below leaves thin coverage of ordinary transformation performance. Report those results descriptively. General performance and indistinguishability claims require a larger, separately powered study across writers and registers.

Before live reward use, include adaptive style-metric attacks in that larger study. Give the search procedure the published evaluator and allow repeated edits selected to maximize V under a declared budget. Compare with unchanged sources and ordinary editing, with matched generation/compute budgets for competing revision methods, on held-out source families and writers. Retain query histories and all candidates; report metric gains, blinded author-fit judgments, semantic failures and search costs, including disagreement among revisions that pass P, R, T. Freeze the evaluator and rubric for each comparison; any correction enters a later validation round. This study is separate from the 24-task protocol pilot.

Within that pilot, reserve four B source/brief comparisons for chosen-candidate semantic attacks: ambiguous attribution or negation, ambiguous preservation of qualifications, borderline readability, and disputed compliance with tone constraints. Each comparison uses three committed candidate variants: an intended clear pass, a clear violation, and an attempted unresolved judgment. Include a requested dramatic tonal shift as an intended pass, and certainty changes, unrequested tone reversal and loss of required informality among the violations. Briefs without separate tone instructions still require target-tone certification, using the requested authorial style or the source-tone default as appropriate. Full meaning preservation remains mandatory in every case. Freeze the rubric, validator weights, reviewer assignment and candidate budget before generation. Review all variants independently under the same field rules; an unresolved variant must not erase a certified competitor's result. Keep every assigned slot in aggregation. This gives twelve candidate evaluations within the fixed pilot budget, separate from chosen-input execution-failure controls.

Record field-level `PASS`, `FAIL` and abstaining weight, candidate-specific unresolved rates, shared-failure scope, review time, candidate/report costs and retained competitor credit. Replay the frozen allocation with and without each attack entry, keeping assigned weights, to measure a simulated common owner's share when sibling UIDs score elsewhere. Test individual withholding under deliberate and benign ambiguity, and shared-scope closure under infrastructure failures. Test selective validator non-certification and attempts to misclassify a candidate-triggered error as shared evidence failure. Publish these counterfactuals as diagnostics. This bounded study cannot establish resistance to all adaptive attacks or native payout manipulation.

In parallel, use synthetic paid B jobs and test funds to verify instance-key input encryption, customer-only result delivery and exactly one settlement. Require valid receipt plus complete on-chain ciphertext plus a timely strict weighted certificate. Test payment with no customer acknowledgment; fake, stale, replayed or wrong-model receipts; substituted input/output keys; missing, partial or altered ciphertext; duplicate or insufficient signer weight; and certificate-only attempts to bypass contract checks. Test every cutoff equality and refund/payment race, certificate delay after valid delivery, revoked-at-acceptance keys, later policy updates and changed instances. Invalid uploads must not block a valid completion, and any relayer must preserve the fixed payee. Keep certification timeouts distinct from proven execution failure. Neither validators nor gateways receive plaintext or keys. Measure gas, finalized-history retrieval and verification capacity for 4, 16 and 64 KiB padded result envelopes before freezing the maximum. Ordinary A acknowledgment remains a separate hosting mode.

Use the writer pilot to establish acquisition and review costs. Refit public starter models only on data cleared for the intended release. Evaluate author and origin models on new sources, then train the transformation adapter after enough reviewed pairs exist. Reward scaling requires demonstrated judge reliability and preservation performance, with uncertainty reported on unseen sources.

The one-ticket rewrite mailbox exercise in Section 10 is a transport and attribution pilot. It does not supply the complete common three-A validator-execution matrix or qualify every UID for emissions. A full tournament must reserve its expanded traffic and global review workload under a separately frozen schedule before issue. The 24-task offline tournament is not a demonstrated 360-block throughput claim.

Before a live subnet launch, resolve the supported Pangram retrieval/billing integration, key and assignment registry, replay ledger, public A artifact qualification and reference execution, protocol-derived counterpart and reserve assignments, service-ticket authorization and consumption, measured B rewrite-service quotas, delivery-dispute attribution, fixed measurement, evidence, review, and weight deadlines, and complete review coverage for every scored submission, empty-pool weight policy, and mechanism allocations. Exercise exhausted quotas, duplicate tickets, concealed common control, strategic refusal, and same-artifact runner recovery. Verify the cached panel and thresholds freeze before B task disclosure; no post-issue detector substitution is permitted. Verify that execution failure cannot become B nonresponse, an automatic pass or a private request's reassignment. Validate the greater-than-10% failure gate in aggregate and on each mandatory service interface and assigned role, with exactly 10% passing and zero workload supplying no automatic eligibility. Test requester abandonment, registration changes, deterministic probation, and minimum fallback response windows. Subnet UID trimming can renumber surviving UIDs [37]; pause new protocol issue during such a migration and verify the lifetime-slot ledger, registration bindings and outstanding deficits before resuming. Native renumbering must not silently reset protocol counters or rewrite issued assignments. Demonstrate the zero-credit rule and persistent recovery deficits, including same-class probation and resistance to resets through key or service-version changes. Verify settlement under native weight-reveal delays and payout cycles; already distributed emissions cannot be clawed back by this proposal. Before paid customer launch, separately establish bounded certification-timeout exposure, retention terms, encrypted delivery reliability, and alpha escrow behavior under finality and timeouts. Verify paid-roster state proofs, frozen security policy and revocation collateral, full ciphertext checks and verification funding before accepting a job. Verify quote expiry, capacity reservations, exact alpha accounting, release and refund permissions, and fee handling. These requirements must be concrete protocol parameters or measured acceptance criteria before funds depend on them. Reproduce the frozen style scores, field certificates, individual withholding and shared-scope decisions from archived inputs. Demonstrate proof-verified validator snapshots, complete mailbox and witness publication, fault attribution and finality handling. Verify B self-scores through independent validator execution. Demonstrate complete artifact availability, canonical content identity, reference-runtime agreement, independently labeled qualification and targeted-trigger controls. Test pending A commitment copying, wrong registration bindings, late or invalid openings and missing artifacts. Verify that only a valid opening and successful qualification preserve the original finalized commitment's priority, and that copying a newly disclosed artifact cannot outrank that entry. Include already-public artifacts and confirm that registration credit establishes no authorship claim. Test exact-copy coalescing, near-copy incentives, chosen-input execution failures and full-validator replay cost. B's local public-detector feedback must be available during the pilot; measure transfer to fresh source families, later A versions and Pangram.

Before either B emissions or paid B launch, demonstrate attested inference with the private starter model on a pinned CPU/GPU configuration. Independently review the client verifier, reproducible runner, measured policy, private model loader and complete input/output path. Test host and guest administration, fake or stale quotes, wrong nonces, unrelated GPU evidence, unprotected offload, debug mode, key substitution, revoked firmware, model replacement after attestation, malicious model data and an endpoint that signs text it did not generate. Validate the same private version's benchmark receipts and customer receipts without exposing its weights.

Use synthetic market-sensitive announcements to test denied egress, plaintext logs and crash dumps, hostile output markup or remote-resource fetches in the customer client, cache and state isolation, output-key binding, metadata leakage, restart/rollback, replay and revoked instances. Include malicious weights that encode source facts in early stopping, errors or workload shape; verify padded output and fixed release slots under the accepted graph and bucket policy. Test independent model-owner and customer key release, including an owner trying to substitute its own recipient after provisioning. Check that no model-owner key broker receives customer keys and no customer or separate host can retrieve weights through storage, logs or application interfaces. Measure attestation freshness, maximum job duration, startup time, throughput, cost per accepted revision and supported model size before freezing the launch policy. Include customer refusal of a policy update and loss of attestation service availability. No attested request may silently become ordinary hosting; ordinary B customer offers must be rejected. A failed confidential-compute pilot delays B emissions and paid B. Ordinary transport encryption cannot qualify operator confidentiality.

Reject ordinary B submissions from emission allocation. Retain their offline utility only as research diagnostics. Test missing or invalid receipts, private-version copying, post-issue downgrade, external best-of-many selection and changes in the paid serving profile. Repeat the same frozen task across restarts and qualified instances to verify the bound result and execution budget. Test benchmark recognition and measure the registered version on authorized service requests from the paid interface. Required training tickets must use that version too. Attestation adds no quality bonus and does not change semantic review or Pangram obligations.

Measure paid B demand and miner margins separately from emissions, with A hosting as a secondary offer. Include competitors purchasing outputs, model extraction attempts, operator downtime and confidential-hardware concentration. A private model and a TEE provide no guaranteed revenue or permanent evasion advantage.

The next deliverable is a small reproducible tournament and private-job pilot, followed by revised parameters based on observed failures and reader judgments. The whitepaper remains a design draft until those results support deployment.

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